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AI-based drive-by methodology for early damage detection in a steel truss railway bridge under multiple environmental and operational factors

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ABSTRACT

Railway infrastructure monitoring is fundamental for transportation safety, particularly as networks expand and structures age. While traditional approaches rely on direct sensor installation on bridges, these systems are costly and impractical for widespread implementation. Despite advances in drive-by monitoring techniques, existing methods struggle to handle environmental and operational variations (EOVs) effectively, limiting possible applications in real-world conditions. This study develops an AI-based drive-by automated methodology for detecting structural damage in railway bridges. Using a three-dimensional finite element model of a Warren-type truss bridge and vehicle-track interaction simulation, the methodology was tested through comprehensive numerical simulations for freight railway operations. These simulations, offering essential flexibility for developing the technique, constitute a key step before experimental validation. The proposed damage detection framework encompasses multiple stages: initial preprocessing for noise reduction, feature extraction using Continuous Wavelet Transform, deep learning analysis *via* Convolutional Autoencoders, and final damage assessment using Structural Similarity Index error maps combined with statistical outlier detection. Results demonstrate successful structural damage detection across various scenarios, even at early stages, sensitivity to damage severity, and robustness against EOVs. This research provides a cost-effective alternative to traditional railway bridge monitoring systems while maintaining robust damage detection capabilities. Future works include experimental validation.

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1. Introduction

Railway infrastructure systems play a pivotal role in economic development by efficiently supporting passenger and heavy freight transport. However, the natural ageing of these structures requires regular inspections and monitoring to ensure structural integrity and the optimal performance of their components. In the Brazilian context, railways are primarily utilised for transporting goods, such as iron ore, characterised by low-speed operations and heavy loads. This demand has driven plans to expand the railway network, with proposals for new routes already announced by the National Agency of Terrestrial Transportation (ANTT). Notable projects include Ferrogrão, the North-South Railway, the West-East Integration Railway, the Rumo Malha-Norte Railway, and the New Transnordestina Railway (Holler Branco et al., 2022). This expansion drives a proportional increase in the demand for bridge structural health monitoring (SHM), making anomaly detection a critical task for risk mitigation (Carnevale et al., 2019; Meixedo et al., 2021).

The most straightforward approach to damage detection involves installing sensors directly on the structure. However, this method faces financial barriers due to the costs of implementing individualised and real-time monitoring systems. Typically, monitoring systems with multiple sensors require traffic interruptions during their installation, frequent maintenance of their components, and a continuous power supply. As a result, their implementation is often limited to a select minority of critical bridges (Malekjafarian et al., 2022). To address the limitations of direct damage detection methodologies, emerging interest in drive-by techniques has gained momentum since the pioneering work of Yang et al. (2004). This study demonstrated that vibration data from a vehicle passing over a bridge contain information about the bridge's dynamic properties. The proposed method eliminates the need for sensor installation on the structure, as the vehicle itself becomes the instrumented system while also acting as a dynamic loading actuator. Consequently, a single vehicle can be used to monitor multiple bridges (Souza et al., 2023).

Despite the potential of drive-by monitoring, implementing methodologies that perform effectively under real conditions requires strategies to distinguish changes in features due to damage from those resulting from Environmental

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and Operational Variations (EOVs). As a rule, the more sensitive a feature is to damage, the more it is also sensitive to EOVs (Farrar & Worden, 2012). Several authors, including Bernardini et al. (2021), Li et al. (2023), and Zhang et al. (2023), highlight the challenges and the need to enable drive-by systems for real-world environmental conditions, such as traffic interference, environmental noise, seasonal temperature variations, wind, track irregularities, and other traffic influences.

The removal of EOVs from the data is typically known as a data normalisation process. This operation is greatly facilitated when it is possible to measure and model environmental actions; however, this is not always feasible in SHM systems (Farrar et al., 2001) and is even more challenging in a drive-by monitoring context. A traditional method for removing EOVs is the use of Principal Component Analysis (PCA). By modelling the features in a principal component space and removing those with greater variability, considered inherent to EOVs, authors have successfully managed to handle EOVs (Meixedo et al., 2021). Although these methodologies often succeed, they frequently encounter difficulties in effectively normalising more severe EOVs. Consequently, damage indicators that previously performed well under less challenging conditions begin to exhibit deficiencies in more complex scenarios (Silva et al., 2023).

In the drive-by context, the impact of EOVs has been addressed either through parametric analysis—where the sensitivity of the detection method to different variables is studied—or holistically, through the verification of the methodology's robustness to identify damage regardless of the operational scenario. However, the existing literature has largely overlooked variables such as minor manufacturing defects (e.g. geometric or material imperfections), thermal effects, and vehicle positioning accuracy. Instead, research has predominantly focused on the influence of track irregularities, measurement noise, and vehicle properties such as travelling speed (Malekjafarian et al., 2022; Souza et al., 2023).

Considering the current state of the art in drive-by monitoring for detecting damage in railway bridges, this work aims to provide innovative contributions on aspects that are not thoroughly detailed in the existing literature, namely:

- Developing a comprehensive and automated methodology for damage detection in railway bridges utilising unsupervised learning, with a focus on application in freight vehicles, a research domain that is still insufficiently explored (Malekjafarian et al., 2022; Souza et al., 2023) with existing contributions still heavily focused on high-speed passenger trains (de Souza et al., 2025) or dedicated to supervised approaches (Hajializadeh, 2022a, 2022b).
- Evaluate the robustness of the methodology in a problem considering various EOVs inherent to the operating environment of a freight railway, such as vehicle localisation errors, wagon mass variations, measurement noises, temperature-induced elasticity modulus variations, irregularity profiles, and minor insignificant damages in the bridge, accounting for possible fabrication tolerances and defects.
- Evaluation of the methodology's performance using more complex numerical models of the bridge and vehicles, both experimentally calibrated. Most works in the field use non-realistic scaled experiments or highly simplified numerical models, where the bridge is often treated as a simply supported beam element. To the best of the authors' knowledge, only a few authors have worked with more complex bridge models (Bernardini et al., 2021, 2025).
- Explore the performance of the methodology in the detection of more realistic cases of degradation of isolated secondary elements (e.g. bracings) that are more prone to damage (Haghani et al., 2012), distinguishing from previous works in the area which express damage as a uniform stiffness loss across the entire cross-section of the bridge deck (Souza et al., 2023).

This study represents a critical step in advancing the current state of the art in drive-by methodologies for damage detection, which largely remain limited to overly simplified numerical simulations. By leveraging the flexibility of numerical studies, this research explores complex scenarios that would be challenging to replicate experimentally at this stage. The insights gained lay the groundwork for transitioning towards experimental validation, ensuring that the methodology is robust and ready to address the complexities of real-world applications. As a part of a broader research initiative, this study focuses on refining the methodology itself, while future works will aim to validate its effectiveness in experimental settings. This phased approach balances the need for rigorous development with the eventual goal of practical implementation. To highlight the contribution of this work, Section 2 gives a broad overview of the state of the art on drive-by bridge monitoring.

2. Background on drive-by bridge monitoring

Since the introduction of the method, drive-by techniques have been explored by several authors to extract the dynamic properties of bridges, including natural frequencies, vibration modes and damping coefficients, fomenting modal-based damage detection techniques. However, these have shown to be more sensible to environmental and operational conditions (EOVs) than damage itself (Malekjafarian et al., 2022). Modal-based techniques limitations led authors to seek alternative damage features, mostly by employing advanced signal processing techniques, such as time-frequency analysis. These analyses have the advantage of offering temporal resolution and handling non-stationary signals, which is usually the case in SHM. The most traditional approaches for extracting features in the time-frequency domain include Hilbert transforms, short-time Fourier transforms (STFT), and continuous wavelet transforms (CWT).

Leveraging the capabilities of CWT, McGetrick and Kim (2013) proposed a method based on the peak values in the differences between wavelet coefficients from healthy and damaged states to identify and localise damage in a bridge. Their methodology proved to be more effective for longer span bridges and at lower vehicle speeds. Hester and González (2017) corroborated these findings and concluded that the damage detection efficiency is compromised when the damage-induced frequencies are too close to the frequencies of other sources of vibrations, emphasising that heavier vehicles could yield more robust results due to the lower relative dynamic-static effects. Building on this premise, Demirlioglu and Erduran (2024) proposed a methodology that consists of selecting a frequency region at which the CWT scalogram is not disturbed by bridge and vehicle natural frequencies or by the driving frequency. Their technique performed satisfactorily for low to moderate road roughness levels and stiffness loss of 20% or greater. However, this approach may experience a loss of reliability for damage-induced frequencies nearing the bridge's first natural frequency, and the boundary effects inherent to the CWT were not mitigated, resulting in signal cropping near the supports.

The fundamental concept of contrasting wavelet coefficients between baseline and damaged states was also applied by Fitzgerald et al. (2019), who processed vehicle acceleration signals to detect bridge scour, modelled as a reduction of the vertical foundation stiffness. A distinct approach was adopted by Tan et al. (2020) who combined CWT with the theory of Shannon entropy rather than utilising wavelet coefficients directly. Their method identifies the wavelet function scale that fits the signal in the baseline scenario with the lowest entropy and assumes that the wavelet function at this specific scale will be sensitive to signal disorder, thus enabling the damage detection.

While most of the applications of CWT has been explored for beam bridges, Bernardini et al. (2021) investigated the potential of detecting damaged connections in trussed bridges using Hilbert-Huang and CWT. In their study, realistic cases of localised damage were modelled as a reduction in diagonal elements' stiffness, and a case of bolt's looseness in the bottom chord connection was modelled through an equivalent spring substituting the damaged connection. Changes in the wavelet energy peak were also explored in the work of Tan et al. (2022) to estimate bridge conditions, while damage localisation was conducted using the component wavelet energy. Recent advances have also focused on refining the input signal itself. Moving beyond the vehicle's response, the work of Zhang et al. (2023) proposed using the characteristic wavelet coefficient as a damage indicator, derived by processing the contact-point response using CWT and complete ensemble empirical mode decomposition with adaptive noise.

In addition to the requisite for features that are sensitive to structural response variations caused by damage, any SHM methodology to perform under real conditions requires effective strategies to distinguish changes in features due to damage from those resulting from EOVs. With the advancement of machine learning methodologies, many researchers consider neural networks a significant breakthrough for modelling and subsequently eliminating the effects of EOVs in complex scenarios (Locke et al., 2020). A particularly promising type of neural network is the autoencoder. This network aims to reconstruct an original representation (decoder) based on a compressed representation (bottleneck) created by an encoder, thereby forming the autoencoder. In the context of SHM, these networks can serve both as feature extractors, where the compressed representation is utilised as sensitive features (Lee et al., 2022), and as normalisation elements. In the latter case, the networks are trained exclusively on undamaged scenarios, and their reconstruction error is employed as a normalised feature that is sensitive to damage but not to EOVs (Sarwar & Cantero, 2021). Studies specifically focused on drive-by applications have already demonstrated superiority of autoencoders compared to PCA (Calderon Hurtado et al., 2023; Liu et al., 2020).

Although promising, applications of autoencoders in a drive-by context remain scarce, with most studies focusing on road vehicles (Calderon Hurtado et al., 2023; Li et al., 2023; Liu et al., 2020; Sarwar & Cantero, 2021). Calderon Hurtado et al. (2023) used an adversarial autoencoder to improve residual spectrum response reconstruction in two-axle road vehicles, outperforming PCA and traditional autoencoders. Liu et al. (2020) identified a nonlinear link between road vehicle acceleration and bridge damage severity, with significant data at different frequencies, showing stacked autoencoders beating PCA in feature extraction. Sarwar and Cantero (2021) proposed a damage index for road bridge monitoring based on the Kullback-Leibler (KL) divergence for the reconstruction error distribution of deep autoencoders (DAEs) computed for several vehicles crossing the bridge. Based on numerical simulations, it was demonstrated that the method can detect damage under operational conditions. In the scope of time-frequency analysis, Li et al. (2023) applied an unsupervised DAE model in a laboratory experimental study, where the STFT was utilised to assess bridge conditions. The squared prediction error of the autoencoder, trained on baseline scenarios, was used as the damage metric.

While this body of research illustrates a growing interest in autoencoders for road applications, the railway domain remains comparatively underexplored. To the best of the authors' knowledge, only three papers addressed railway bridges to date (Bernardini et al., 2025; de Souza et al., 2025; Fernandes et al., 2024). de Souza et al. (2025) employed sparse autoencoders to reconstruct log mel-cepstral coefficients and proposed an indicator based on the KL divergence of the mean absolute reconstruction error. This methodology, validated through simulations on 3D models for high-speed trains, proved to be robust and effective in detecting damage in simply supported bridges. The feasibility of using the log mel-cepstral coefficients as damage features was established in previous work by de Souza et al. (2022), which also concluded that these coefficients exhibit high sensitivity to speed and track irregularities,

demonstrating the need of the autoencoders and the KL divergence. Complementing these advances, Fernandes et al. (2024) applied DAE to acceleration signals simulated in passenger vehicles for the detection of structural anomalies in bridge functions. The proposed methodology involved training autoencoders on acceleration signals corresponding to undamaged scenarios and using the mean absolute error as a damage-sensitive feature. Indicators for a series of passages were calculated, and the KL divergence was employed as a damage indicator to distinguish the error distribution of a baseline scenario from a damaged one. Furthermore, the authors proposed a scheme based on Receiver Operating Characteristic analysis to optimise the positioning of sensors on the vehicle.

In the domain of time-frequency analysis, Bernardini et al. (2025) utilised Sparse Autoencoders and wavelet coefficients to detect damage in a truss bridge subjected to corrosion, modelled as material mass loss and stiffness reduction. Vertical acceleration data from a single sensor installed in the front bogie of a passenger train were processed utilising CWT and truncated according to the frequency range of interest, governed by the modular configuration of the bridge as well as the forward speed of the train. The resulting wavelet coefficients were then averaged for each instant in time, and the resulting curve was used as input in the Sparce Autoencoder. The damage indices were obtained from the mean absolute error and T^2 Hotelling's statistic.

Vehicle-based systems have also been identified experimentally as a promising avenue to enhance railroad service availability and reliability. Rakoczy et al. (2016) evaluated the feasibility of employing a track deflection measurement system installed in a lightweight passenger car, as well as both track deflection and geometry measurement systems mounted on an operational track measurement vehicle, to detect weak stringers and pier elevation changes in an open-deck trestle bridge. More recently, Rakoczy et al. (2024) conducted an exploratory study on service bridges in Poland and concluded that proper processing of onboard track geometry measurements can yield valuable information on bridge condition. The study further emphasised that railway companies could benefit from establishing baseline scenarios of bridges across the network to support the implementation of damage detection software. The findings underscored the potential of such approaches and emphasised the need for further research in this area. To illustrate the contribution of this work, Table 1 summarises the key aspects covered by the literature mentioned in this section.

3. Damage identification methodology

This section provides a comprehensive overview of the proposed damage identification methodology, which is based on data collected from a vehicle operating over the bridge. The methodology consists of four distinct stages: (i) data acquisition and pre-processing; (ii) feature extraction; (iii) feature fusion; and (iv) feature discrimination.

3.1. Data acquisition and pre-processing

Initially, acceleration data is collected by various sensors distributed along the vehicle. The sensors should be installed at strategic positions on the vehicle, typically one sensor per structural component (axles, bogies, and carbody) for better performance of the methodology. A clear rationale for the selection of these points is discussed for the case study of this article in Section 4.2.

Once collected, this data is pre-processed to enhance feature extraction and machine learning algorithm training. First, the data is detrended, low pass filtered to remove spurious noise, and decimated to reduce the computational complexity of subsequent operations. The appropriate selection of the filter cut-off frequency and decimation ratio is crucial for the methodology's success. Essentially, it must be ensured that the portion of the spectrum relevant to the bridge's dynamics is preserved, and non-relevant frequencies are eliminated, as they tend to increase the complexity of the methodologies without adding value. Furthermore, the vehicle responses are then separated by velocity classes, which is easily achievable in practice, with tachographs or GPS, ensuring a higher accuracy without compromising practical implementation.

3.2. Feature extraction

Feature extraction process is composed of three primary steps. First, Wavelet coefficients are extracted from the acceleration data. Next, the coefficients associated with intact scenarios are used to train deep convolutional autoencoder (DCAE) models. Finally, the trained autoencoders are used to predict the Wavelet coefficients of each scenario, and Structural Similarity Index (SSIM) Error Maps are computed as damage-sensitive features.

3.2.1. Continuous wavelet transform

The first step in feature extraction involves the application of the Continuous Wavelet Transform (CWT) to the pre-processed acceleration signals. The CWT is a powerful signal processing tool that provides a suitable time-frequency representation for the analysis of non-stationary signals. The CWT tiles the time-frequency plane with variable-sized windows, where the window widens in time for low-frequency phenomena and narrows for high-frequency phenomena. This makes it suitable for analysing transient behaviour, rapidly changing frequencies, and slowly varying behaviour.

In essence, the CWT decomposes a signal into a set of translated and scaled wavelet functions, derived from a so-called mother Wavelet function. The mathematical definition of the CWT of a signal $x(t)$, for a scale a and time shift b is given by:

$$CWT_w(a,b) = \frac{1}{|a|^{\frac{1}{2}}} \int_{-\infty}^{\infty} x(t)\overline{\psi}(t)\left(\frac{t-b}{a}\right) \mathrm{d}t \qquad (1)$$

in which t stands for time and $\overline{\psi}(t)$ is the complex conjugate of a particular mother wavelet function in time domain (Olhede & Walden, 2002).

Table 1. Overview of key aspects considered in the drive-by bridge monitoring literature mentioned in this section.

Reference	Type of analysis	Type of vehicle	Type of bridge	EOVs*
McGetrick and Kim (2013)	Parametric	Half-car model/Experimental setup	Beam	1, 2, 3, 7, 8
Hester and González (2017)	Parametric	Half-car model	Beam	1, 2, 6, 7, 8
Fitzgerald et al. (2019)	Holistic	Train carriage calibrated model	Beam	1, 2, 3, 6, 7, 8
Liu et al. (2020)	Parametric	Single-degree-of-freedom sprung mass model/Experimental setup	Beam	1, 7
Locke et al. (2020)	Holistic	Highway quarter-car model convoy	Beam	1, 2, 3, 4, 7, 8
Tan et al. (2020)	Parametric	Quarter-car model/ Half car model	Beam	1, 6, 7, 8
Bernardini et al. (2021)	Parametric	Two models of passenger train: two-bogie carbody and shared bogies between adjacent carbodies	Truss	1, 2, 6, 7, 8
Sarwar and Cantero (2021)	Holistic	5-axle truck model	Beam	1, 2, 3, 6, 7, 8
Tan et al. (2022)	Parametric	Quarter-car model	Beam	1, 2, 3, 6, 7, 8
Zhang et al. (2023)	Parametric	Single-degree-of-freedom sprung mass model	Beam	1, 6, 7, 8
Calderon Hurtado et al. (2023)	Holistic	Half-car model/Experimental setup	Beam	1, 2, 6, 7, 8
Li et al. (2023)	Parametric	Experimental truck	Beam	1, 2, 6, 7
de Souza et al. (2022, 2025)	Parametric	Four-axle car high speed train model based on Siemens ICE3 Velaro	Beam	1, 2, 6, 7, 8
Fernandes et al. (2024)	Holistic	Passenger train calibrated model	Beam	1, 2, 3, 6, 7
Demirlioglu and Erduran (2024)	Parametric	Half-car model	Beam	1, 2, 7, 8
Bernardini et al. (2025)	Holistic	Passenger train calibrated model	Truss	1, 2, 3, 5, 6, 7, 8
This work	Holistic	Freight-train calibrated model	Truss	1, 2, 3, 4, 5, 6, 7, 8, 9

*EOVs description: (1) Irregularity profile/road roughness, (2) Vehicle speeds, (3) Changes in vehicle properties, (4) Temperature effects, (5) Positioning accuracy, (6) Measurement noise, (7) Damage severities, (8) Different types/location of damage, (9) Manufacturing defects.

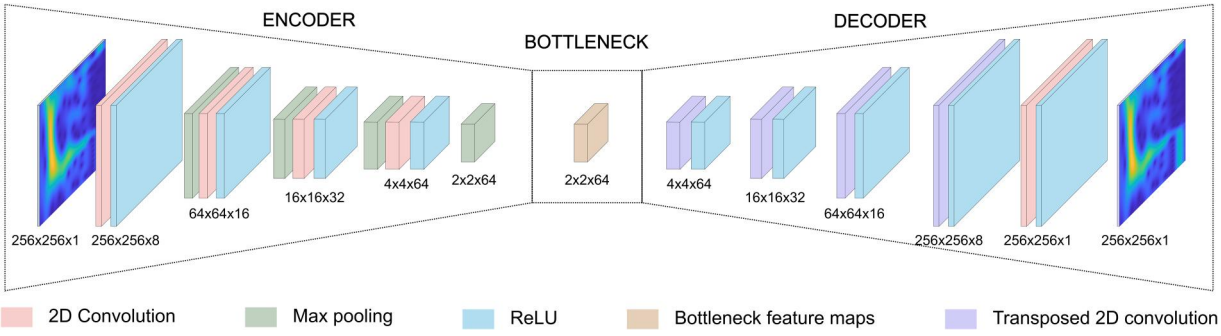


Figure 1. Graphical representation of the CAEs architecture. From left to right: the input scalograms are encoded by sequences of convolution, ReLU and max pooling layers and decoded by transposed convolutions and ReLU layers to achieve a reconstructed version.

In the case of this work, a Morse mother wavelet function ($\Psi_{\beta,\gamma}$) was used, which is defined in frequency domain by:

$$\Psi_{\beta,\gamma}(\omega) = U(\omega) a_{\beta,\gamma} \omega^{\beta} e^{-\omega^{\gamma}} \tag{2}$$

in which $a_{\beta,\gamma} \equiv 2\left(\frac{e\gamma}{\beta}\right)^{\beta/\gamma}$ is a normalising constant, $U(\omega)$ is the Heaviside step function, γ is a symmetry parameter, and β is a decay parameter (Lilly & Olhede, 2009). In this study γ is assumed to be equal to 3 which gives perfectly symmetric wavelets and β equal to 20 which gives a 60-sample time bandwidth. This mother wavelet was selected because it outperformed both Bump and Morlet wavelets for this task. Additionally, the ability to adjust the temporal and frequency resolutions using the parameters γ and β provides greater flexibility for adapting the methodology to other scenarios. In the calculations, 48 voices per octave were used for better frequency resolution and a frequency range that encompasses the bridge main vibration modes.

It is also important to highlight that due to the translation of the mother wavelet, boundary coefficients are imprecise. For lower frequencies, this imprecision extends further in time, forming a cone of influence. These coefficients were discarded in this study and do not compromise the bridge signal, as they refer to small portions of track before and after the bridge.

3.2.2. Convolutional autoencoders

The second step in the feature extraction process involves training CAE for each sensor to learn how to reconstruct the Wavelet scalograms. Since the CAEs are trained exclusively on data from normal operational conditions, it is assumed that they will not adequately handle atypical and unseen situations, such as bridge damage. Therefore, the CAEs perform a feature normalisation, which aims to remove the influence of EOVs. This network handles these EOVs, as it learns to manage these scenarios considered normal during training. When using the network's prediction error as a feature, it is already normalised.

3.2.2.1. Network architecture. The architecture of the CAEs, depicted in Figure 1, was designed semi-empirically, balancing the authors' experience with testing. The encoder uses a combination of convolutional, ReLU, and max pooling layers (pink, blue, and green layers in Figure 1).

Convolutional filters extract relevant features, the ReLU layers introduce non-linearities, and max pooling performs dimensionality reduction and compression, as well as improving the network robustness to minor data shifts. The decoder mirrors this with transposed convolution layers (purple layer in Figure 1) for up sampling and ReLU layers to learn non-linear combinations for better image reconstruction. A final convolutional combines the deconvolved filter maps into a final reconstructed image. Additionally, the implementation of a ReLU layer before the output is strategical to ensure the absence of negative pixel, which are incompatible with the reconstruction of absolute wavelet coefficients.

Layers were arranged to achieve an aggressive 4× reduction initially, with a gentler 2× reduction at the bottleneck, and the number of filters was progressively increased in powers of two. This configuration was chosen to achieve a fast compression at the more general and bigger feature maps and a refinement at the more abstract smaller and specific ones. Finally, the number of filters and their size were determined through testing to ensure a balance between computational efficiency and accurate scalogram reconstruction, avoiding an oversized network that would merely copy inputs to the output.

Complementing the visual representation provided by Figure 1, specific details regarding each layer are presented in Table 2 for reproducibility. Still regarding the proposed architecture, it is worth emphasising that the input images are compressed by a factor of 256, which is a consequence of number of filters and bottleneck feature maps size. This compression is essential to ensure that the network learns the system's underlying dynamics rather than merely reproducing the input images.

Finally, to deal with the fixed size input of the CAE architecture, the input scalogram images are resized by nearest-neighbour interpolation to a uniform dimension (256×256) to ensure the same network structure and compression ratio can be used for different passages. CNN architectures rely on convolutional operations, which apply filters or kernels to local regions of an input image. Standardising input dimensions into a square format simplifies these operations, ensuring that the filters can efficiently traverse the entire image without the complications associated with uneven dimensions.

3.2.2.2. *Structural similarity index-based loss function.*

Besides the definition of the architecture, a custom loss function based on the SSIM was designed. The SSIM is a metric introduced by Wang et al. (2004) to evaluate the similarity between two images, x and y, considering aspects such as luminance, contrast, and structure. The SSIM metric yields a number between zero and one, reflecting increasing similarity between the images, and is formally defined as:

$$\mathrm{SSIM}(x,y) = [l(x,y)]^{\alpha} \cdot [c(x,y)]^{\beta} \cdot [s(x,y)]^{\gamma} \tag{3}$$

where α, β, and γ are exponents that allow for assigning distinct weights to each of the components of the SSIM. In the specific case of this work, an equal weight was adopted for each of these parameters ($\alpha = \beta = \gamma = 1$). The terms l, c, and s are related to luminance, contrast, and structure given by:

$$\begin{aligned} l(x,y) &= \frac{2\mu_x\mu_y + C_1}{\mu_x^2 + \mu_y^2 + C_1} \\ c(x,y) &= \frac{2\sigma_x\sigma_y + C_2}{\sigma_x^2 + \sigma_y^2 + C_2} \\ s(x,y) &= \frac{\sigma_{xy} + C_3}{\sigma_x\sigma_y + C_3} \end{aligned} \tag{4}$$

Table 3. Properties of the wagon numerical model (Bragança et al., 2022).

Parameter	Symbol (unit)	Calibrated value
Carbody		
Mass	m_{cb} (ton)	45.72
Moment of inertia about longitudinal axis	$I_{cb,x}$ (t.m^2)	49
Moment of inertia about transverse axis	$I_{cb,y}$ (t.m^2)	562.8
Moment of inertia about vertical axis	$I_{cb,z}$ (t.m^2)	665
Length	L_{cb} (m)	10.000
Width	W_{cb} (m)	2.170
Height above ground	H_{cb} (m)	2.297
Wheelsets		
Mass	m_w (kg)	1247
Moment of inertia about longitudinal axis	$I_{w,x}$ (kg.m^2)	312
Moment of inertia about vertical axis	$I_{w,z}$ (kg.m^2)	312
Track gauge	W_{cp} (m)	1.668
Height above ground (wheelset)	H_{ws} (m)	0.450
Suspensions		
Longitudinal stiffness	$k_{1,x}$ (kN/m)	44981
Lateral stiffness	$k_{1,y}$ (kN/m)	30948
Vertical stiffness	$k_{1,z}$ (kN/m)	2252.7
Vertical damping	$c_{1,z}$ (kN.s/m)	35.73

Table 2. Details of the configuration adopted for each layer in the CAE.

Layer/type	Component role	Filters	Filter/pool size	Padding/cropping	Stride	Activation
imageInputLayer (z-score normalisation)	Encoder	—	—	—	—	—
convolution2dLayer		8	9	Same	—	ReLU
maxPooling2dLayer		—	4 × 4	—	4 × 4	—
convolution2dLayer		16	9	Same	—	ReLU
maxPooling2dLayer		—	4 × 4	—	4 × 4	—
convolution2dLayer		32	9	Same	—	ReLU
maxPooling2dLayer		—	4 × 4	—	4 × 4	—
convolution2dLayer		64	9	Same	—	ReLU
maxPooling2dLayer		—	2 × 2	—	2 × 2	—
transposedConv2dLayer	Decoder	64	9	Same	2 × 2	ReLU
transposedConv2dLayer		32	9	Same	4 × 4	ReLU
transposedConv2dLayer		16	9	Same	4 × 4	ReLU
transposedConv2dLayer		8	9	Same	4 × 4	ReLU
convolution2dLayer (Output)		1	9	Same	—	ReLU

in which μ, σ, and σ_{xy} are the mean, standard, and cross-covariance and C are regularisation terms given by:

$$\begin{aligned} C_1 &= (0.01L)^2 \\ C_2 &= (0.03L)^2 \\ C_3 &= \frac{C_2}{2} \end{aligned} \tag{5}$$

in which L is computed as the absolute difference between the maximum and minimum pixel values of the image.

Equation (3) is then computed for each pixel of the image considering an 11×11 square region around it. The adjacent pixel values are weighted by a Gaussian function with a standard deviation of 1.5, meaning that the further away from the zone, the lesser the contribution to the index. Later, an average can be taken to provide a global quality index ($SSIM_{mean}$), which is used in this step to construct the loss function. Considering that the perfect similarity would have $SSIM_{mean}$ value equal to 1, the error of reconstruction is calculated by taking the absolute value of one minus the $SSIM_{mean}$ value. Thus, the loss function used for training the CAE was defined as:

$$\text{Loss} = \sqrt{\frac{1}{N}\sum_{j=1}^{N}\left(1 - SSIM_{meanj}\right)^2} \tag{6}$$

in which N is the batch size used for training. Compared to metrics such as mean absolute error or mean squared error, which are more frequently used, SSIM provides a more comprehensive assessment of the image by considering additional aspects beyond merely the numerical difference in pixel intensity at corresponding positions.

3.2.2.3. Training of the deep convolutional autoencoder model. From the architecture defined in Section 3.2.2.1 and the loss function defined in Section 3.2.2.2, the final model can be finally trained with the undamaged scalograms to receive an unknown input and reconstruct it after encoder-decoder transformation. In the specific case study explored in this work, the training details of the model are presented in Section 4.8. From a computational perspective, training these autoencoders is the costliest part of the entire methodology but is done only once during the methodology, since it is conducted only in the baseline scenarios (undamaged bridge). However, in a real operational case, this operation might need to be performed occasionally to increase robustness as more data is collected for the intact bridge. Additionally, the acquisition of baseline data and the subsequent training of the convolutional autoencoders must be performed for each bridge, creating a specific bank of autoencoders for each bridge in the inventory to be monitored.

3.2.3. Structural similarity index error maps

The third step in feature extraction involves using the DCAE trained according to Section 3.2.2 to reconstruct the input scalograms and from them compute the SSIM map for each sample. In this case, the SSIM value, calculated using Equation (3), is computed for each pixel. To ensure that representative error values are higher, 1 minus the SSIM map is computed, meaning that a value of zero indicates total similarity, while a value of 1 indicates poor representation. After this operation, a time-frequency map of the reconstruction error of the CAEs is finally obtained. These maps serve as damage-sensitive features and are used as the basis for constructing a unique damage indicator.

3.3. Feature fusion

In this stage, the features, i.e. the various pixels of the SSIM maps, are combined to enhance their sensitivity to damage and produce a unique indicator that allows for an easy evaluation of the bridge's integrity. First, the average error along the temporal dimension of the map when the vehicle is over the bridge is taken, preserving changes that are consistent over the time the vehicle crosses the bridge. This approach was selected considering that in a continuous structure formed by several elements like a bridge, a loss of stiffness in one of them will result in a change in the overall behaviour pattern of the structure, thus impacting the vehicle's response during its passage and even some moments after it has crossed. Although local changes also occur and are often preferred in traditional SHM damage identification schemes, it is unlikely that they can be detected by drive-by methodologies. This is because the scanning is only conducted on the deck and would not capture, for example, local effects in a particular element of a truss bridge.

Once the averaging is done, the features now only have the frequency domain dimension. In this case, it is not beneficial to use an averaging strategy, considering that errors at different frequencies may indicate distinct or varying severity damages. Therefore, each of these frequency values is assumed to be an element of a multivariate statistical distribution, and the squared value of the Mahalanobis distance (MD) is used as a parameter for data fusion. This metric allows evaluating the distance of a point from a statistical distribution, which in this case is assumed to be the undamaged bridge. The MD can be computed by:

$$MD = \sqrt{(\mathbf{X} - \boldsymbol{\mu})^T \sum^{-1} (\mathbf{X} - \boldsymbol{\mu})} \tag{7}$$

in which $\mathbf{X}$ denotes the vector of the features of a particular observation, $\boldsymbol{\mu}$ is the mean vector of the feature multivariate distribution, and $\sum$ is the correspondent covariance matrix (Ghorbani, 2019).

Finally, after this process, a damage index is calculated for each train passage and for each sensor. The last step then involves the fusion of these indices from the sensors into a single sensitive indicator. For this, the squared value of the MD is applied once again, resulting in a unique sensitive indicator.

3.4. Feature discrimination

After calculating the damage indicators, the final step is the automatic detection of damage. For this purpose, a statistical

threshold (TH) was calculated from the reference scenario of the DIs, based on the Gaussian cumulative inverse distribution:

$$TH = invF_x(1-\alpha) \tag{8}$$

in which

$$F(x|\overline{\mu},\sigma) = \frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{x} e^{-\frac{1}{2}\left(\frac{y-\overline{\mu}}{\sigma}\right)^2} dy \tag{9}$$

and σ is the standard deviation, $\overline{\mu}$ is the mean and α is the significance factor. In the scope of this work, the TH was computed for a confidence level of 99.99% to avoid false negative detections. This choice was based on the premise that not reporting a fault is more severe than occasionally issuing false alerts. The methodology discussed in this section and its steps are summarised in the flowchart presented in Figure 2, which integrates the four stages into a single schematic view.

4. Numerical case study

This section provides a thorough overview of the numerical methodology employed for the validation of the drive-by damage identification methodology described in Section 3. Initially, the numerical models of the three principal railway subsystems - vehicle, track, and bridge - are introduced. Subsequently, detailed insights into the numerical vehicle-structure interaction simulations are provided, along with a comprehensive examination of the various scenarios investigated in the study.

4.1. Freight wagon numerical model

The vehicle is a two-axle Laagrss-type flat wagon (Figure 3(a)), featuring a single level suspension system that connects its axles to the carbody *via* a parabolic leaf spring with variable stiffness. This component provides suspension stiffness through the flexure of its leaves and damping through the friction between them. The springs are secured using UIC double links (UIC, 2007). The axles are spaced 10 m apart, and the total length of the vehicle is 14.8 m. It has an empty weight of 27.1 tonnes and a cargo capacity of up to 24.9 tonnes.

The vehicle was modelled in ANSYS Inc (2018) using a rigid body approach (Figure 3(b)), wherein all structural components were treated as rigid, with only the suspensions considered flexible modelled by COMBIN14 elements. The inertial properties of the vehicle were assigned at the centres of gravity for both the carbody and wheel-axle assemblies, through MASS21 elements. This modelling technique strikes an effective balance between accurately capturing the overall dynamics of the vehicle and achieving the computational efficiency necessary to conduct the extensive simulations required for training machine learning models (Bragança et al., 2022). The properties adopted for modelling the vehicle are presented in Table 3 and were calibrated and validated through direct comparison between numerical and experimental dynamic records by Bragança et al. (2022).

4.2. Virtual monitoring system

A virtual monitoring system, incorporating eight vertical acceleration sensors, whose positions are depicted in Figure 4, was used. Four sensors were installed at the axle level (purple triangles in Figure 4), with one sensor placed on each axle box, and four sensors were positioned at the carbody level (red triangles in Figure 4), directly above the suspensions. Installing sensors on the axles is a strategic decision, as these components are the closest to the bridge that can be instrumented, and collecting data from both sides facilitates the detection of potential asymmetries in the bridge's behaviour caused by twisting. Instrumenting various points at the ends of the carbody is also strategic; it enables the capture of both vertical translation movements and rotational movements of this element by considering the differential movements between its components. These positions were validated as feasible for the installation of this type of sensor during the validation campaign of the numerical wagon model presented in Bragança et al. (2022).

4.3. Bridge numerical model

The bridge is a Warren-type steel truss with a span of 21.42 m, a width of 4.5 m, a height of 3.71 m, and a first bending frequency of 16.0 Hz and first torsion of 19.5 Hz. The model was developed utilising data available in the literature (Bernardini et al., 2021) and calibrated to match the experimental frequencies. The complete structure was modelled in ANSYS Inc (2018) using BEAM188 elements. Figure 5 provides a visual representation of this model from multiple perspectives, including top and lateral views. The mechanical properties of the Warren truss bridge elements are presented in Table 4. Due to limited information, the bridge's damping ratio was assumed to be 0.5%, based on recommendations from EN 1991-2 (2003). The numerical model used Rayleigh damping matrix (Chopra, 1995), fixing the 0.5% ratio for the first two modes.

4.4. Track numerical model

The modelling of the track followed a layered approach based on the model proposed by Neto et al. (2021). This model comprises three layers representing the ballast, sleepers, and rails, which are interconnected through appropriate interfaces. The foundation level of the model is fully fixed, while the longitudinal stiffness of the track is provided by the springs' longitudinal stiffness at the different interfaces. In the sections of the track over the bridge, the last two layers corresponding to the ballast and foundation interface are removed, and the sleepers are directly fixed to the bridge's stringers, considering that this is a ballastless bridge configuration. Figure 6 presents an overview of the track model near the bridge, as well as schematic representations of this model in the regions of the track over the foundation and the track over the bridge.

The properties of the numerical model for the track were obtained from the literature and are presented in Table 5.

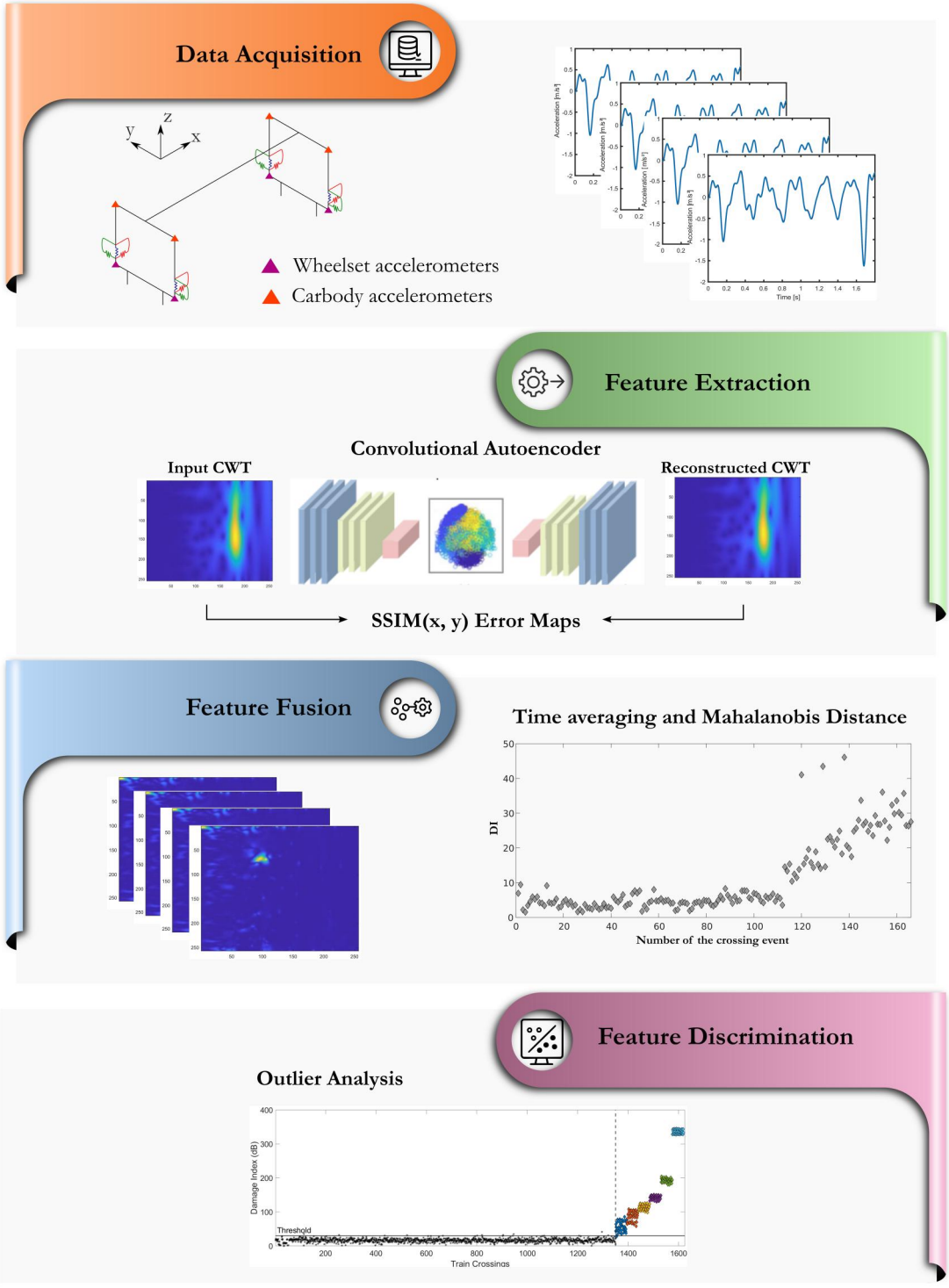


Figure 2. Flowchart of the proposed methodology. Graphical illustration of the main operation involved in each step of the methodology.

Parameter nomenclature is consistent with the one used in Figure 6. To more accurately simulate real-world conditions in track modelling, the simulations incorporate artificially generated railway irregularity profiles derived from power spectral density functions. These profiles encompass wavelengths ranging from 1 m to 75 m with a resolution of 0.01 m. The irregularities were integrated into the simulations using the VSI numerical interaction tool, as described in Section 4.5.

4.5. Simulation scenarios

The simulations considered various scenarios, including both baseline and damaged conditions. Several EOVs were incorporated, such as speed variations, irregularity profiles, wagon mass, temperature fluctuations, wagon positioning errors, measurement noise, and elements with negligible damage (<0.5%). To the best of the authors' knowledge, no previous studies have accounted for such a comprehensive range of EOVs.

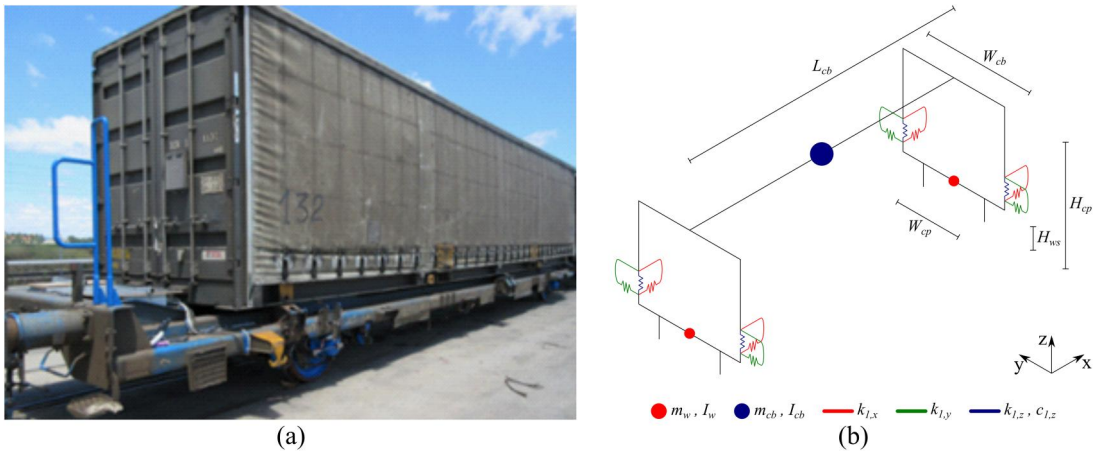


Figure 3. Laagrss-type freight wagon: (a) wagon view and (b) numerical model.

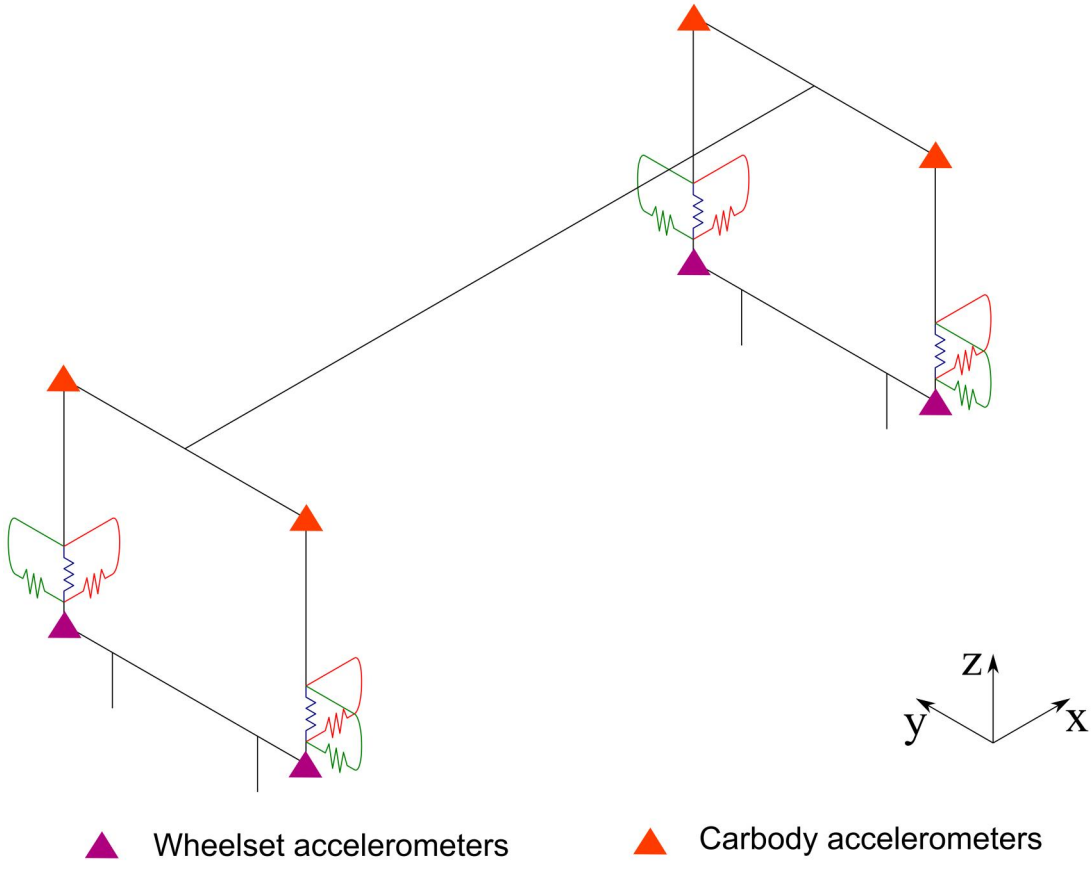


Figure 4. Virtual monitoring system positioning in the numerical model. Wheelset and carbody accelerometers represented by purple and red triangles respectively.

To account for elements with negligible damage, seven samples were randomly generated for each combination of EOVs with damage values below 0.5%. The effects of temperature fluctuations were simplified by changes in the elastic modulus of steel. Considering the bridge's globally isostatic (simply supported) configuration, which minimises these stresses, this simplification was performed to reduce computation complexity required to account for thermal prestress effects. However, more advanced studies are needed to better explore the performance under these effects. Positioning error was considered by randomly varying the starting position of the train in the simulations by ±1 m. The distinct profiles of irregularities introduced into the baseline scenarios comprise the same level of severity, which is consistent with the operational reality of railways, as operators must maintain track irregularities at the same level of severity to preserve the operational class of the track. However, within that level of severity, these irregularities vary over time, both with wear and after maintenance, and the methodology must be capable of differing this from damage.

As indicated in Figure 7, damage was introduced into five distinct elements of the truss bridge, specifically one diagonal member (D1), one of the lower chords of the truss

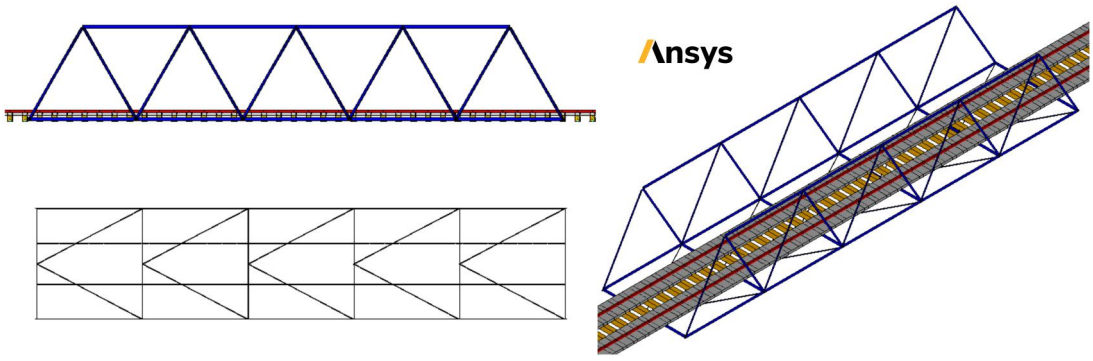


Figure 5. Bridge numerical model. Side, top, and perspective views of the numerical model developed in ANSYS®.

Table 4. Mechanical properties of the bridge numerical model (Bernardini et al., 2021).

Element	Section area A (m^2)	Torsion constant I_t (m^4)	Principal moment I_2 (m^4)	Principal moment I_3 (m^4)
Lower chord	0.0143	1.54×10^{-6}	2.12×10^{-4}	1.8×10^{-4}
Upper chord	0.0148	1.91×10^{-6}	2.14×10^{-4}	1.4×10^{-4}
Side diagonal	0.00615	9.2×10^{-7}	1.65×10^{-4}	4.1×10^{-5}
Bracings	0.0013	4.6×10^{-7}	8.2×10^{-5}	2.1×10^{-5}
Cross-girder (floor beam)	0.0167	6.6×10^{-7}	8.97×10^{-4}	1.9×10^{-5}
Longitudinal girders (stringer)	0.0048	6.4×10^{-7}	2.3×10^{-4}	2.2×10^{-5}

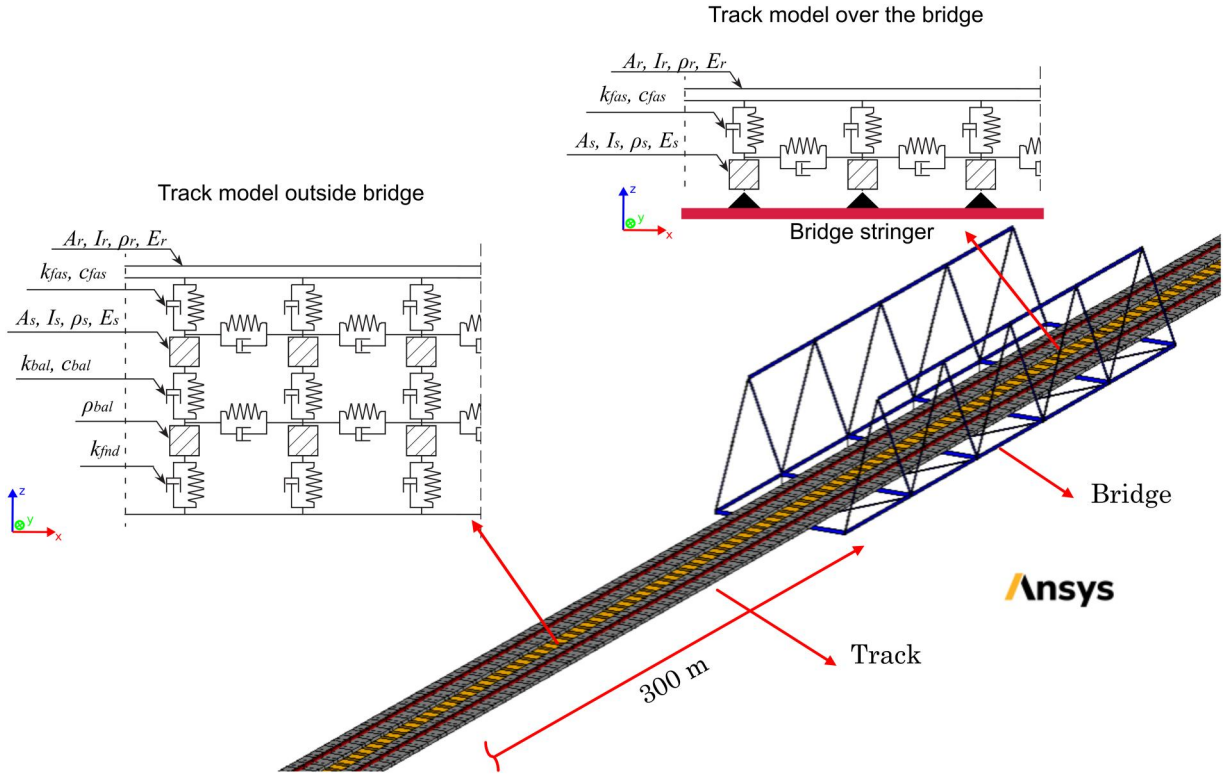


Figure 6. Overview of the track numerical model with schematic representation of the multilayer modelling strategy outside and inside bridge.

(D2), two bracing elements—one in the central area (D3) and another near the support (D4) and a floor beam in the portion between the stringer and the trusses (D5). The simulation of damage was conducted by reducing the stiffness of the bridge member. This approach simplifies the damage to its overall effect on the bridge's dynamic response, rather than detailing the localised failure. For the five elements targeted in this analysis, damage of varying severities ranging from 1% to 100% were introduced. It is important to emphasise that this represents damage to a single structural element of a truss, rather than to the entire bridge. Minor damage was introduced to assess the methodology's ability to detect, in a realistic scenario, potential losses in the cross-sectional area of these elements caused by corrosion, which would affect their stiffness. On the other hand, more severe damage represents significant corrosion, large cracks, or connection failures.

Lastly, the ultimate 100% damage level was introduced to simulate a complete rupture of the element. Ruptures of secondary elements, such as bracings, are relatively common in this type of bridge, and their detection can take considerable time depending on the frequency of asset inspections. It is

Table 5. Properties of the track numerical model.

Component	Parameter	Description	Value	References
Rail (UIC60)	A_r	Area (cm^2)	76,7	EN 13674-1, 2006
	I_r	Inertia (cm^4)	3038	
	ρ_r	Density (kg/m^3)	7850	
	E_r	Elastic modulus (GPa)	210	
Clamps/Pads	$k_{fas,l/t}$	Longitudinal/transverse stiffness (MN/m)	20	Zhai et al., 2009
	$c_{fas,l/t}$	Longitudinal/transverse damping (kN.s/m)	50	
	$k_{fas,v}$	Vertical stiffness (MN/m)	500	ERRI D 214, 1999
	$c_{fas,v}$	Vertical damping (kN.s/m)	200	
	$k_{fas,r}$	Rotational stiffness (kN.m/rad)	45	ERRI D 202, 1999
Sleepers	A_s	Area (cm^2)	402,5	Wu & Yang, 2003
	I_s	Inertia (cm^4)	17620	
	ρ_s	Density (kg/m^3)	2500	
	E_s	Elastic modulus (GPa)	40,9	
Ballast	$k_{bal,l}$	Longitudinal stiffness (MN/m/m)	30	UIC, 2001
	$k_{bal,t}$	Transverse stiffness (MN/m/m)	7,5	ERRI D 202, 1999
	$k_{bal,v}$	Vertical stiffness (MN/m/m)	100	
	c_{bal}	Damping in all directions (kN.s/m/m)	50	Wu & Yang, 2003
	ρ_{bal}	Density (kg/m^3)	1995,9	
Foundation	$k_{fnd,v}$	Vertical stiffness (MN/m)	20	Neto et al., 2021

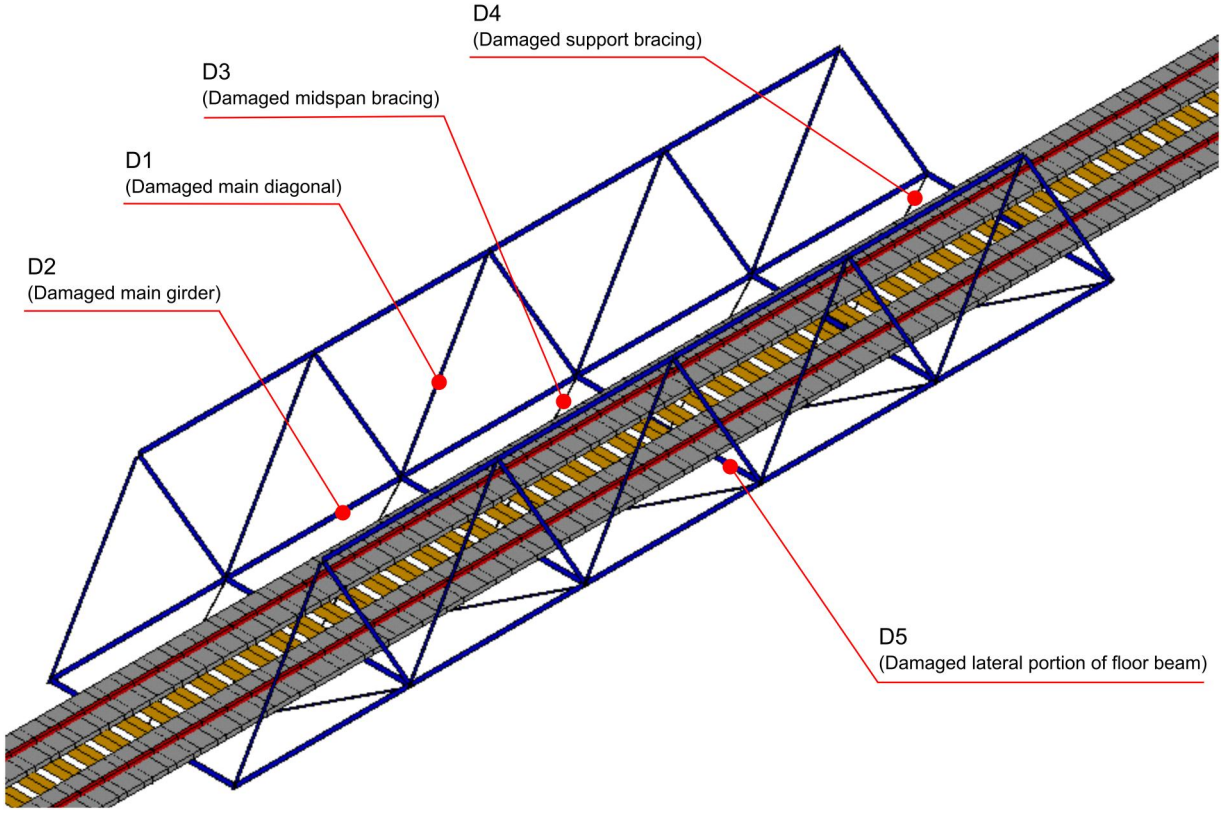


Figure 7. Bridge model with the indication of the five damaged elements.

also worth noting that such damage does not necessarily indicate a complete rupture but could also signify yielding of the steel element. Once the material reaches this yielding stage, there is a sudden decrease in its stiffness; furthermore, the permanent change in member length may lead to looseness and loss of structural function.

The simulation cases considered are condensed in Table 6. The total number of simulations results from the combinatorial analysis, where each variable presented in the table is combined with every case of the other variables, ensuring a holistic approach. In this table, the variations in mass and modulus of elasticity are presented as a fraction relative to the nominal values adopted for the models.

4.6. Vehicle-structure interaction

Comprehensive simulations were conducted across various damage scenarios (see Section 4.5), incorporating the dynamic interaction between the vehicle, the track, and bridge. For this purpose, the in-house numerical tool 'VSI—Vehicle Structure Interaction Analysis' was utilised. Developed by Neves et al. (2014) and later enhanced by Montenegro et al. (2015) to include lateral interactions in the wheel-rail contact model, VSI employs a wheel-rail contact element that connects the vehicle and track subsystems, managing the contact interface by evaluating wheel-rail contact forces in both normal and tangential directions. At each

Table 6. Simulation scenarios: combinatorial EOV test cases.

	Baseline		
Condition	Undamaged	Slightly damaged	Damaged
Number of Laagrss type wagons in the train composition	5	5	5
Speeds (km/h)	45/50/55	45/50/55	45/50/55
Number of irregularity profiles	2	1	1
Wagon mass variation (%)	90/95/100/105/110	90/95/100/105/110	90/95/100/105/110
Change in modulus of elasticity with temperature (%)	97.5/100/102.5	97.5/100/102.5	97.5/100/102.5
Positioning accuracy (m)	±1	±1	±1
Measurement noise (%)	5	5	5
Damage severities (%)	–	<0.5	1/5/10/20/50/100
Individually damaged elements	–	4	5
Number of simulations	90	1260	1350 (5 × 270)

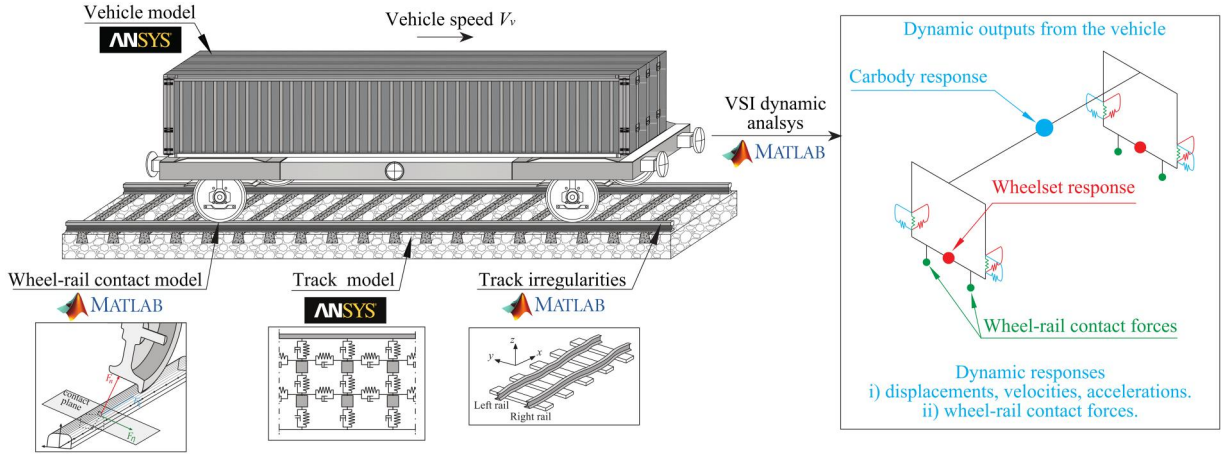


Figure 8. Framework of the VSI numerical tool depicting the integration between different software. Vehicle and track/structure models developed in ANSYS®, while contact model, track irregularities and VSI dynamic analysis performed in MATLAB®.

time step, the algorithm determines the contact point position and calculates the normal contact force using Hertz's nonlinear theory (Hertz, 1882). It also computes lateral and longitudinal tangential creep forces, based on Kalker's rolling contact theory (Kalker, 1979), using the USETAB routine (Kalker, 1996), considering track irregularities. This numerical tool was extensively validated in previous studies considering both established benchmarks and comparison with experimental data (Bragança et al., 2022; Montenegro et al., 2015; Montenegro & Calçada, 2023).

By directly integrating structural matrices from the ANSYS Inc (2018) numerical models of both the vehicle and the structure within the MATLAB® (2020) environment, the tool combines the detailed modelling capabilities of commercial finite element software with the computational efficiency of MATLAB® (2020) programming. This integration enhances the ability to accurately and efficiently simulate complex vehicle-track interactions. Figure 8 provides an overview of the VSI numerical tool framework, graphically showing the interaction between the different software used in the analysis.

4.7. Dynamic responses

After simulation, dynamic responses were detrended and filtered by a low pass 8th order Chebyshev Type I filter with a cut-off frequency of 40 Hz and decimated from 1,000 Hz to an equivalent sampling frequency of 100 Hz. The choice of 40 Hz as cut-off frequency assures that there is no aliasing in the decimation process, and the dynamics of the bridge are properly captured without introducing undesirable frequency components.

Simulated vehicle responses for a speed of 50 km/h, maintaining all other parameters except for damage severity at their nominal values, that is without EOVs, are depicted in Figure 9. These data refer to a sensor positioned at the front of the left side of the carbody and another at the front left axle box. It is evident from the figure that, in the time domain, variations in acceleration signals due to damage are almost imperceptible, except for the damage scenarios D1 and D2 involving element rupture and D5 with 50% of damage. For damages in the bracing elements (D3 and D4), even in scenarios where there is a rupture of this element, the changes in the signal in the time domain are minimal. In the scenario representing the full extent of D5-type damage, in which the rupture of the lateral portion of the floor beam is simulated, the beam behaves as a cantilever connected only on one side of the truss. This condition causes severe deck torsion and subsequent vehicle derailment. Therefore, this scenario is considered unfeasible for analysis and is not presented.

For comparison, Figures 10 and 11 present the envelope of all responses corresponding to the baseline scenario at 50 km/h. Figure 10 considers all EOVs except the

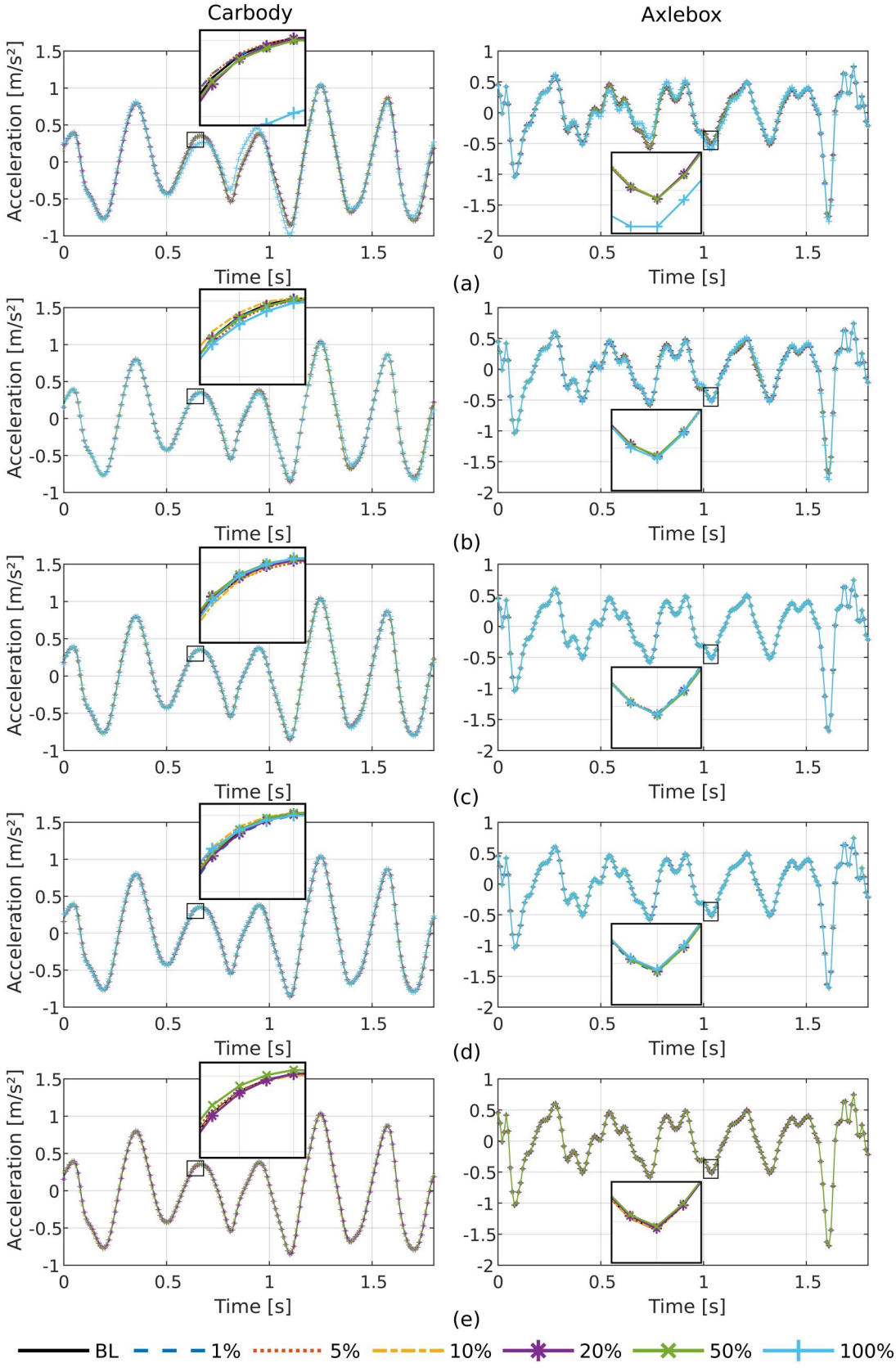


Figure 9. Numerical responses for damage scenarios: (a) D1, (b) D2, (c) D3, (d) D4, and (e) D5.

positioning error, while Figure 11 includes the positioning error and all other EOVs. It becomes evident from this figure that the variations introduced by operational and environmental effects have a much greater magnitude than those caused by the damage scenarios shown in Figure 9, especially when positioning errors are included.

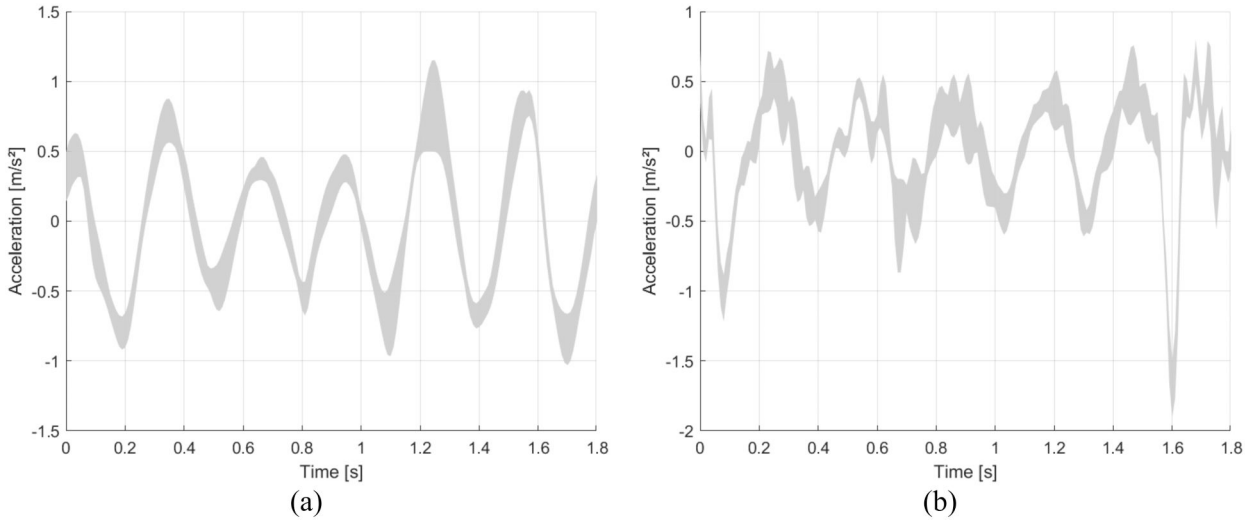


Figure 10. Envelope of baseline responses without positioning error: (a) carbody and (b) axle box.

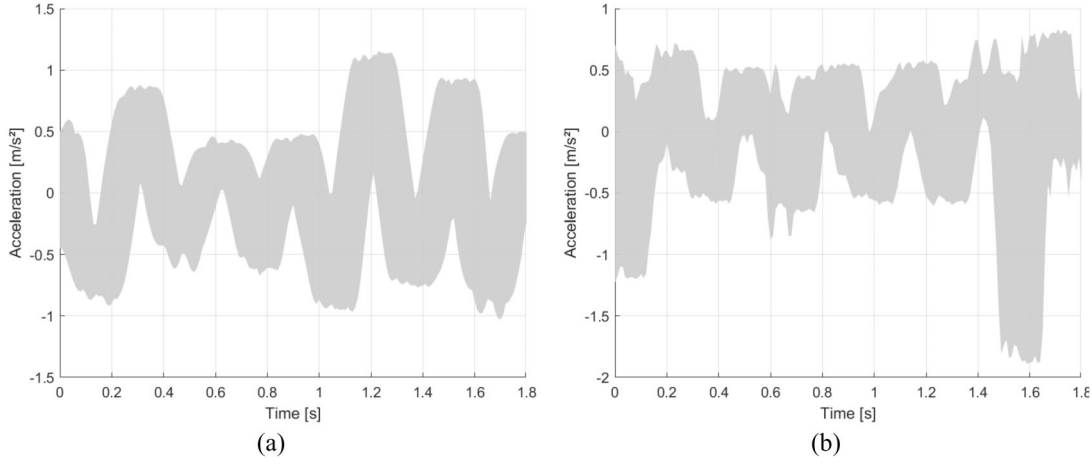


Figure 11. Envelope of baseline responses with positioning error: (a) carbody and (b) axle box.

4.8. Dataset preparation and training strategy

The baseline simulations presented in Table 6 consisted of 450 passages for each of the three speeds considered, totalling 1,350 passages. These were divided into training, validation, and test sets in proportions of 60% (810), 20% (270), and 20% (270), respectively, randomly selected from each speed. Regarding the damaged simulations, all of them are test samples. This training dataset was augmented by applying random translations to the data, generating an additional 810 samples and doubling the training set to 1620, thereby improving its robustness and variability.

The autoencoders whose architecture was described in Section 3.2.2 were trained using the Adaptive Moment Estimation (ADAM) optimiser with a maximum of 500 epochs and an initial learning rate of 0.0001, with early stopping after no significant improvement is achieved in the validation dataset. Validation loss was calculated during training to monitor performance, ensuring the network with the best validation was returned, and training stopped before overfitting. It required 3 h for each autoencoder (one per sensor and speed group) on a computer (Intel Core I5 13600K, 64 GB RAM, Nvidia Quadro P4000 8GB GPU). Figure 12 exemplifies the training curve for one of the CAE. As can be seen, there is no sign of overfitting, and the loss function is significantly improved from 0.790 to 0.010 for the validation dataset and from 0.791 to 0.007 for the training dataset. For all the CAE the best, worst, and average validation losses were 0.003, 0.010, and 0.005 and 0.003, 0.007, and 0.004 for the training dataset, respectively.

5. Results and discussion

In this section, the results obtained from applying the developed methodology to the case study presented in Section 4 are discussed. Initially, partial results related to primary features derived from Wavelet transforms and reconstruction error from autoencoders are presented. Subsequently, the results for each evaluated speed and damage type are discussed.

5.1. Feature extraction

Figure 13 presents the CWT coefficients calculated and predicted by the CAE for a baseline scenario and a damaged

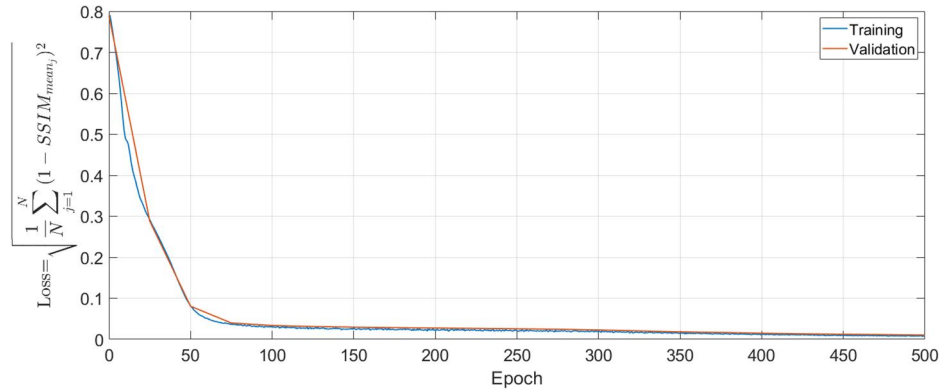


Figure 12. Training/validation history curve of the CAE.

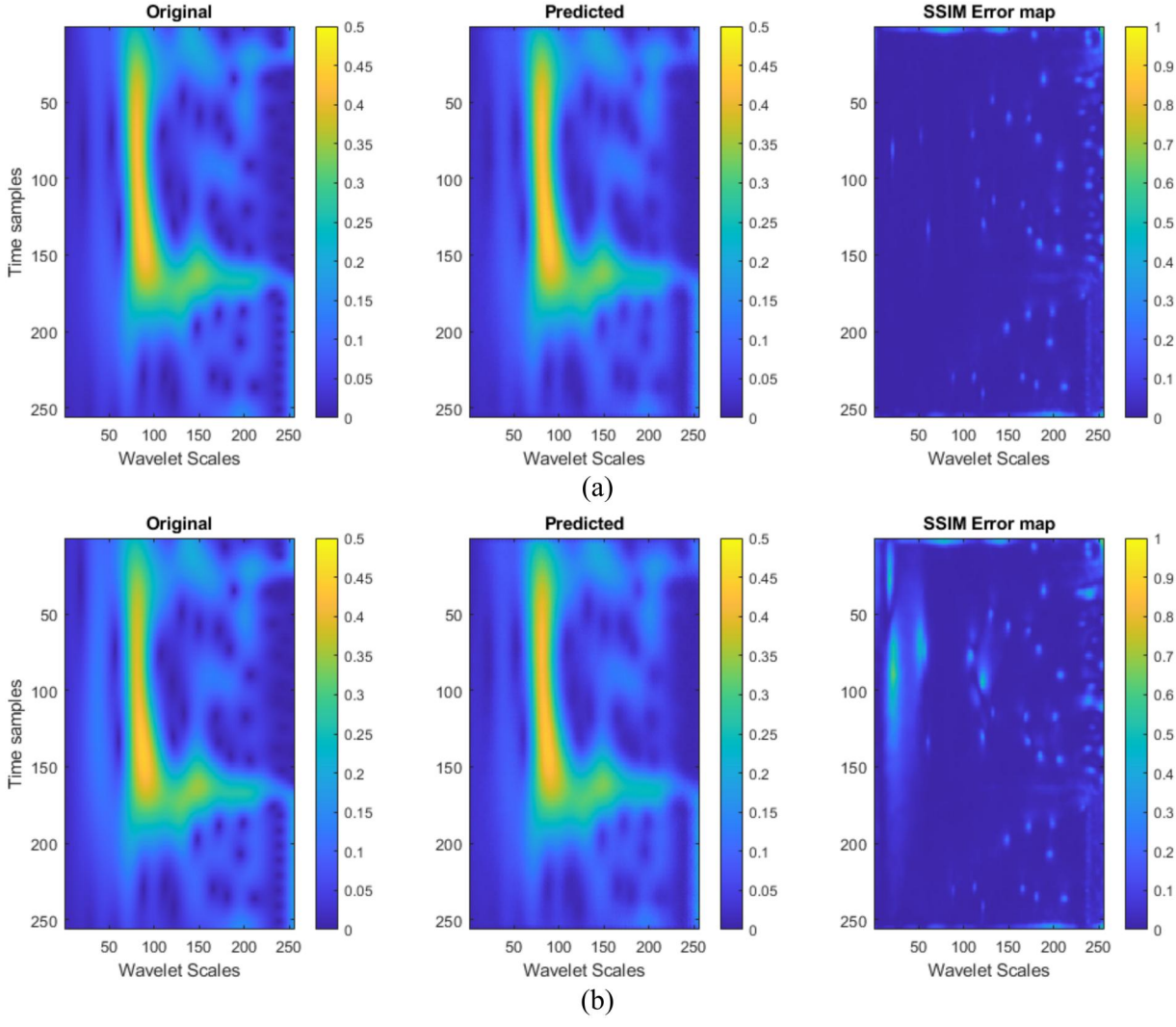


Figure 13. CWT reconstruction performance: (a) baseline scenario and (b) diagonal breakage.

scenario. The data corresponds to a passage at 50 km/h and signal from axle box. The figure shows that the CAE has good performance in predicting scenarios without damage, as evidenced both visually and by the error map. In damage scenarios, however, there is an expected increase in prediction inaccuracies by the CAE, as can be seen especially in the lower Wavelet scales.

5.2. Damage identification

This section presents the results obtained for identifying the various types of simulated damage in the truss bridge. The results are analysed both through the damage index calculated for each individual passage and by utilising boxplots that provide statistical insights into the dataset. In this representation, the median of the data is highlighted, enclosed

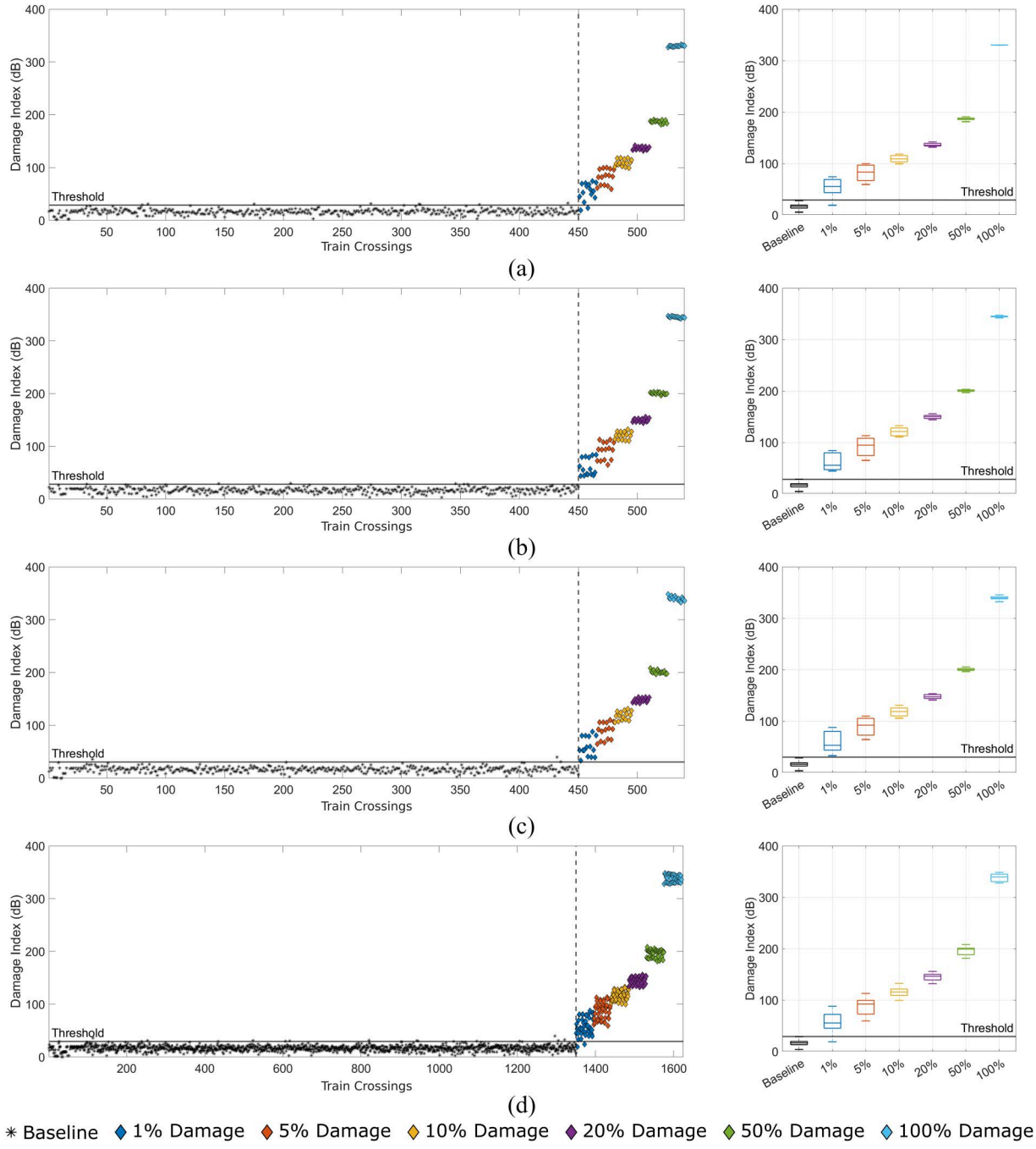


Figure 14. Damage Indices for damaged main diagonal computed for train crossings at: (a) 45 km/h, (b) 50 km/h, (c) 55 km/h, and (d) all speeds combined.

within a small box whose boundaries represent the 25th and 75th percentiles of the dataset. The extreme values are represented by whiskers, while outliers are excluded. The boxplot analysis enables a comprehensive evaluation of multiple passages collectively, facilitating a deeper understanding of the trend and progression of the specific damage under consideration. A more in-depth analysis is conducted for the first case examined, while for the remaining cases, only the specific points of interest relevant to those scenarios are highlighted.

It is important to note that these damage scenarios represent the deterioration of a single truss element rather than the entire bridge. This creates a significantly more challenging scenario for the methodology compared to previous studies in the field of drive-by monitoring, where the entire bridge deck cross-section was subjected to damage with severities as critical as those applied to just one element in this study.

5.2.1. Main diagonal damage D1

In Figures 14(a–c), the results for identifying damage in the diagonal element of the truss at velocities of 45, 50, and 55 km/h, respectively, are presented. Figure 14(d) consolidates the analysis for all three velocities. The scatter plots on the left side depict the damage indexes (DIs) calculated

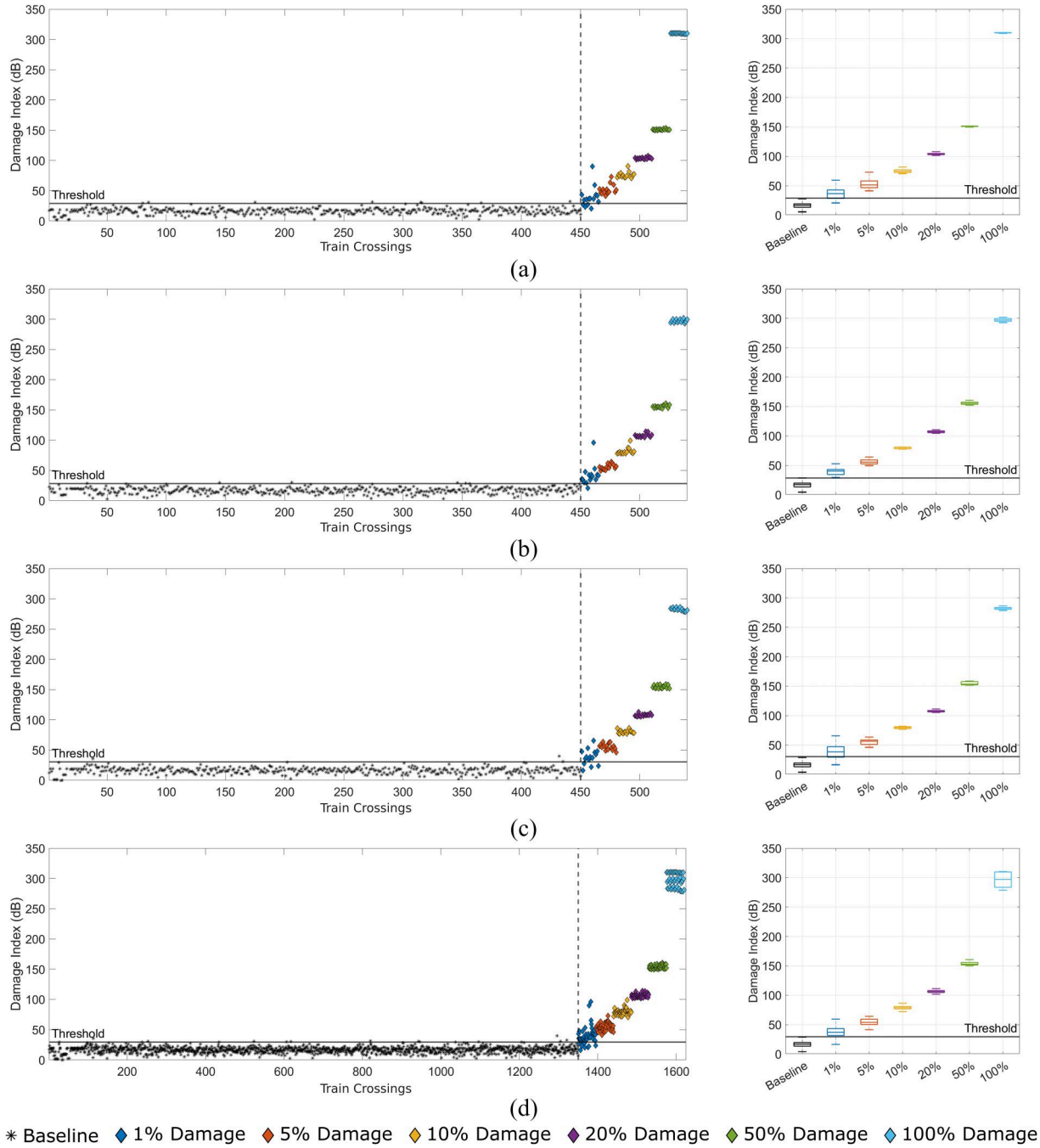


Figure 15. Damage Indices for damaged main girder computed for train crossings at: (a) 45 km/h, (b) 50 km/h, (c) 55 km/h, and (d) all speeds combined.

for each train passage, expressed in decibel scale ($20\log_{10}DI$), given the significant difference in magnitude between the indicator values for varying damage levels.

Upon examining the scatter plots, it becomes evident that the proposed methodology is effective in normalising the EOVs and isolating the effects of the damage. The damage indicators associated with undamaged scenarios (baseline) systematically fall below the detection threshold, with only a few false positives. Additionally, only two false negatives were observed, where damaged data fell below the threshold. These false negatives are associated with scenarios involving 1% damage in a single truss element, which highlights the methodology's high sensitivity. It is also noted that for higher levels of damage, the indicators are more closely clustered, which can be attributed to two primary factors: first, greater damage levels make it easier to distinguish the damage effects from EOVs; second, the logarithmic scale amplifies smaller variations in lower DI values (e.g. around 100) less significantly compared to the same variations in higher DI values (e.g. around 300), which are exponentially larger in a linear scale.

Turning to the boxplots on the right, it is evident that the baseline data consistently falls below the statistical damage detection threshold. Furthermore, there is a clear trend of indicator progression as the damage increases, demonstrating that the methodology can provide insight into

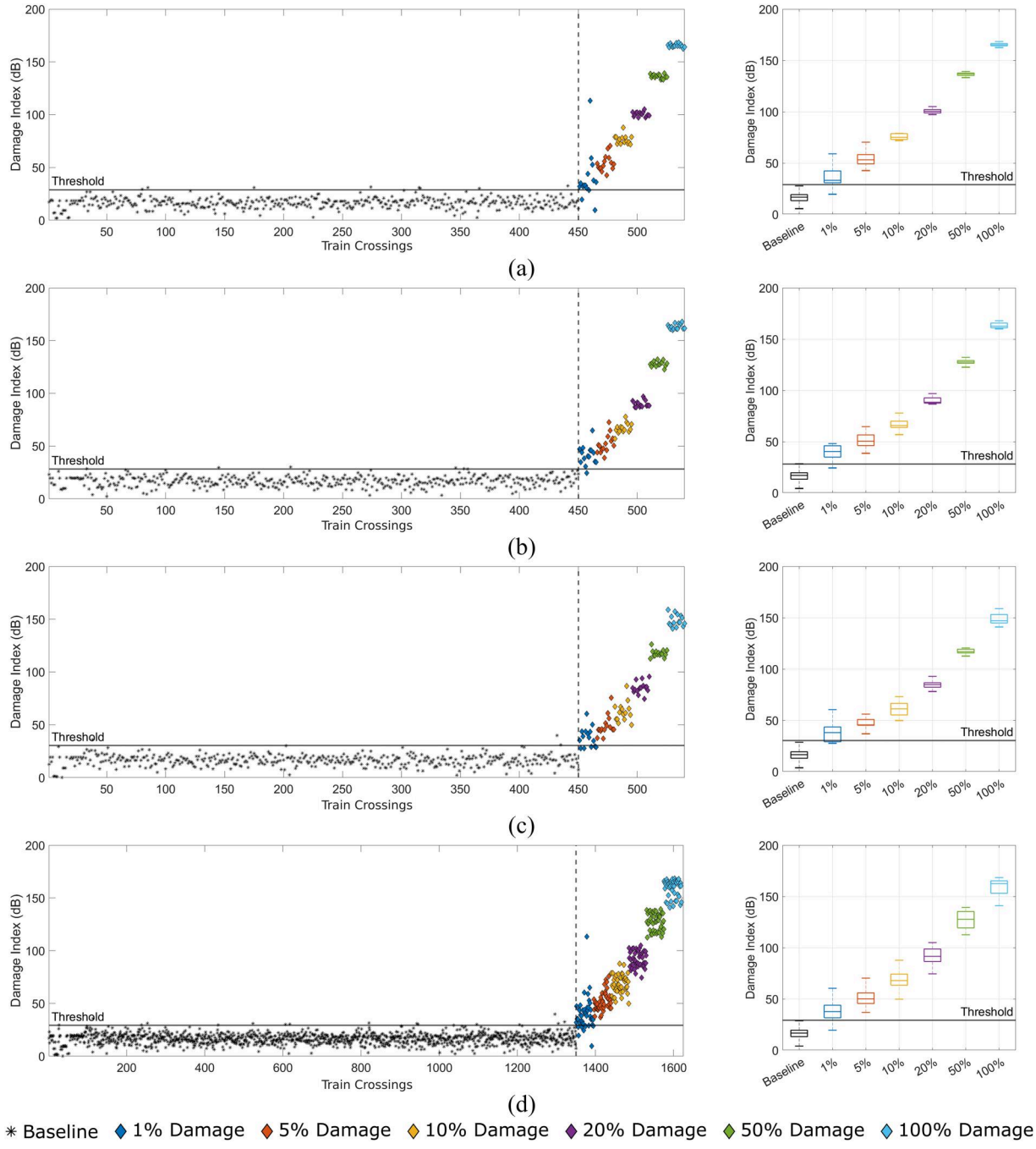


Figure 16. Damage Indices for damaged centre bracing computed for train crossings at: (a) 45 km/h, (b) 50 km/h, (c) 55 km/h, and (d) all speeds combined.

damage progression. This means that, over time, statistical analysis of the data could enable the monitoring of damage progression in the truss bridge elements. Conversely, more severe damage can be identified immediately, without requiring statistical analysis, as such instances can be distinguished indicator-by-indicator without the need for multiple passage analyses.

Finally, it is observed that the results are consistent across the different analysed velocities, demonstrating that velocity is not a hindrance to damage detection. The graphs corresponding to the analysis of all velocities collectively illustrate that damage can be effectively detected, and its severity accurately inferred, regardless of the velocity.

5.2.2. Main girder D2

In the case of damage to the main girder, as demonstrated in Figure 15, the methodology demonstrates its capability to detect damage levels as low as 1%. These detections are statistically superior to both the baseline scenario and the defined statistical threshold, as confirmed through boxplot analysis. Furthermore, the data corresponding to each damage severity is well-clustered, rendering the levels of severity clearly distinguishable, which is also evident from the boxplot.

The presence of some outliers associated with the 1% damage scenario is noted, due to the inherent randomness of certain EOVs and the high sensitivity of the

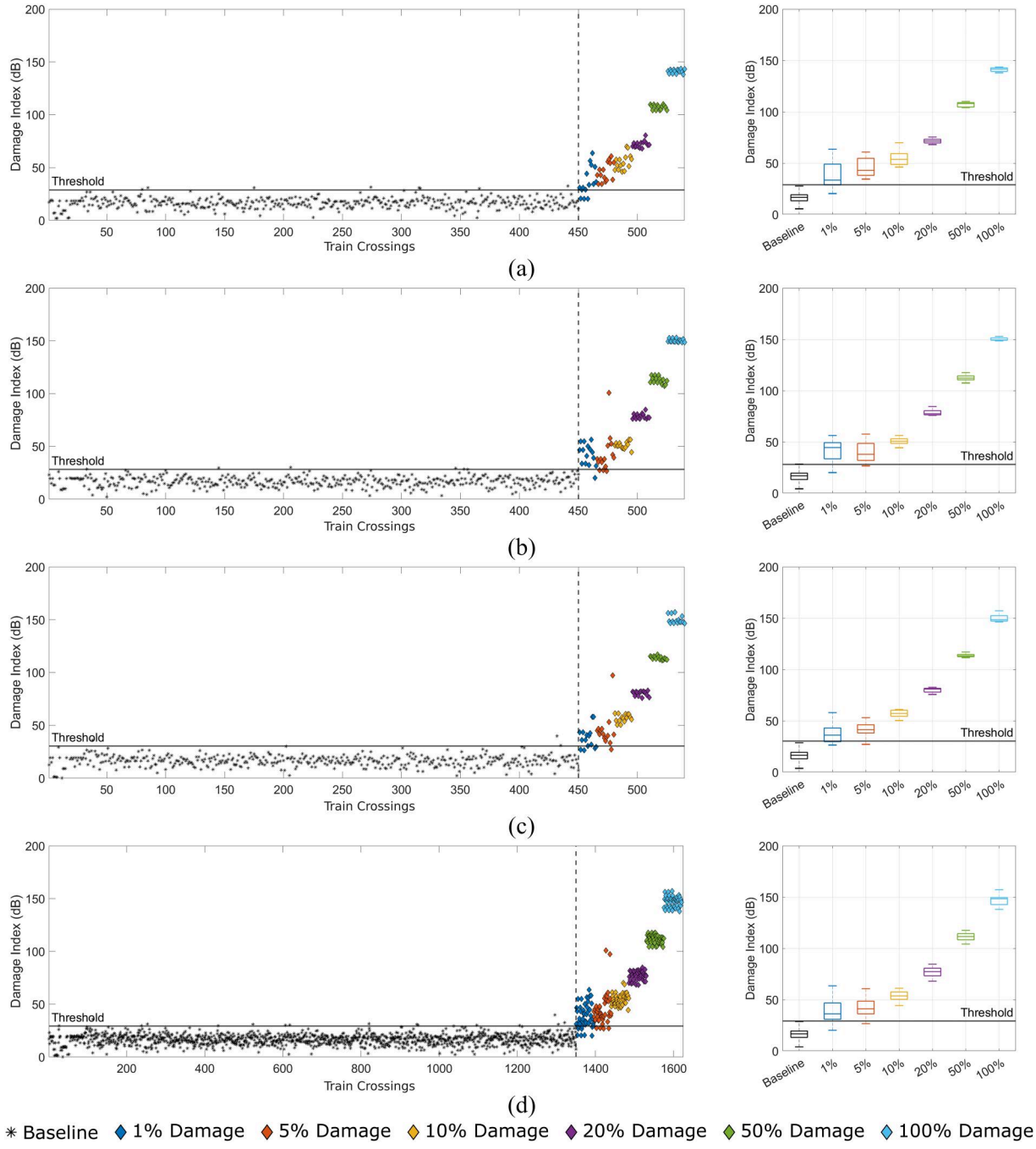


Figure 17. Damage Indices for damaged bracing near the support computed for train crossings at: (a) 45 km/h, (b) 50 km/h, (c) 55 km/h, and (d) all speeds combined.

methodology. However, these outliers can be effectively discarded by conducting a statistical analysis of the remaining data, as presented in the boxplots, ensuring robustness and reliability in the conclusions drawn.

5.2.3. Centre bracing D3

Regarding the damage to the central bracing, as demonstrated in Figure 16, the methodology has proven effective in detecting damage levels as small as 1%, with these levels being statistically superior to both the baseline scenarios and the defined statistical threshold. Sensitivity to the severity of the damage is also demonstrated. Although there is some dispersion for smaller damage levels, it is important to emphasise that even with the rupture of the element, there is minimal alteration in the dynamic response of the vehicle, as demonstrated in Figure 9.

5.2.4. Bracing near the support D4

In damage to the bracing near the support, as presented in Figure 17, the methodology demonstrates its ability to detect damage as small as 1%, while the severity can be effectively assessed for damage levels exceeding 10% in this bracing element. However, this scenario presents significant challenges. Due to the increased dispersion and higher prevalence of outliers, it is not possible to properly distinguish between the damage cases of 1 and 5%. These difficulties

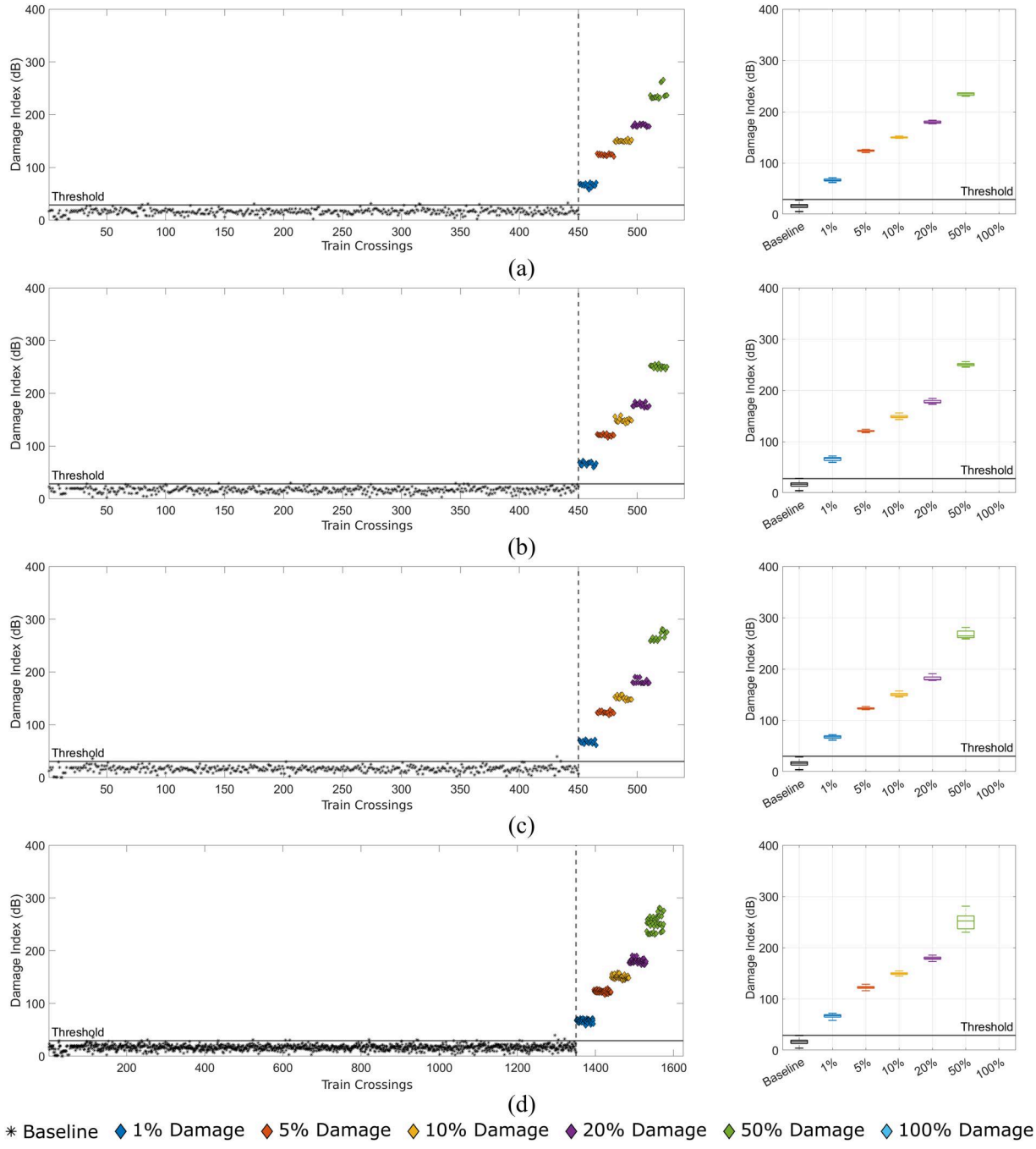


Figure 18. Damage Indices for damaged lateral portion of floor beam computed for train crossings at: (a) 45 km/h, (b) 50 km/h, (c) 55 km/h, and (d) all speeds combined.

arise primarily from the fact that, since this is a secondary element designed to resist torsional loads in a bridge where this type of load is not very significant, the changes in the responses are even smaller than for the other damages resulting in diminutive values of the damage indexes, which hinder the normalisation of EOVs. However, it is worth noting that despite some false negatives, in most cases, the methodology was able to at least identify these damage levels.

5.2.5. Lateral portion of floor beam D5

Finally, regarding the damage at the lateral portion of floor beam, the methodology is shown to handle this type of damage very well, achieving excellent separation between different damage levels and a total absence of false negatives. These results presented in Figure 18 are compatible with the greater severity of this damage and the proximity of the element and vehicle load path.

6. Conclusions

This study presented an AI-based drive-by methodology for detecting structural damage in railway truss bridges subjected to multiple environmental and operational influences. The approach integrates wavelet-based feature extraction with convolutional autoencoders and multivariate statistical

analysis, forming a fully automated framework capable of distinguishing structural changes under realistic and complex conditions. The methodology proved sensitive to different levels of stiffness reduction and demonstrated robustness against a broad spectrum of EOVs, including variations in wagon mass, positioning accuracy, track irregularities, and temperature-induced material changes. Importantly, the framework was tested with freight railway configurations and experimentally calibrated models of vehicle and bridge subsystems, representing a significant step beyond the simplified beam models and passenger train applications commonly found in the literature.

The findings reinforce the potential of vehicle-instrumented systems as cost-effective and scalable alternatives to traditional bridge-based monitoring, which remains limited to a small portion of the railway network due to cost and logistical constraints. By transferring instrumentation to the vehicle, the proposed methodology allows the inspection of multiple bridges using the same platform, providing an efficient strategy to extend monitoring coverage while maintaining high levels of sensitivity to damage.

Nevertheless, the transition to real-world applications poses several challenges, including hardware acquisition costs. Practical deployment must address electronic noise, bandwidth, and storage constraints, and the intrinsic variability of operational excitations such as temperature fluctuations and traffic-induced vibrations. Instrumenting the vehicle requires careful definition of sensor locations, protection of measurement systems, and integration with positioning technologies. In addition, reliable acquisition systems with edge computing and stable power supply are fundamental to ensure data quality in operational environments.

Additionally, it is presumed that the transition to other bridge types, damages in different elements, and varying load conditions will not compromise the methodology's applicability, due to its unsupervised nature and to the absence of a priori assumptions about the type of structure or the damage location in the framework conception. However, conditions in which a bridge is excessively stiff and not sufficiently excited by the vehicle may require some adaptations, such as the frequency range used in the Wavelet transform, which may need to be widened or narrowed depending on the natural frequencies of the bridge in question.

Future research will focus on refining the methodology for damage severity assessment, increasing robustness against more operational and environmental influences, including thermal induced prestress, enhancing the methodology to achieve damage localisation, more realistic damage modelling, and conducting an experimental validation campaign. Most importantly, experimental validation is already underway, with data currently being collected on a damaged metallic bridge and scheduled for continuation after repair. This effort represents a decisive step towards transferring the methodology from simulation to practice, ultimately contributing to the large-scale adoption of drive-by approaches for railway bridge health monitoring.

Declaration of generative AI technologies in the writing process

During the preparation of this work, the authors used Microsoft Copilot® and Google Gemini® to improve language and readability, since English is not the native language of the authors. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Disclosure statement

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